

Sony AMS-100 Audio Monitor Speaker

A Technical Monograph

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Section 1 — Introduction & Historical Context

1.1 Scope of This Monograph

This monograph documents the Sony AMS-100 Audio Monitor Speaker: its design, its modular option-board system, its role in professional broadcast infrastructure, and the surviving units documented by this project. As of the literature sweep of July 2026, no substantive editorial or community documentation of the AMS-100 exists anywhere — no retrospective articles, no forum teardowns, no collector references. The present work is therefore primary literature. Where a claim rests on thin evidence, that is flagged inline rather than smoothed over; where a price is an asking price rather than a confirmed sale, it is identified as such throughout.

The evidentiary core of the work is the author’s own hardware. The author (David, “Orlandu81”) has owned and documented seven AMS-100 frames, which constitute the registry of Section 4; the registry is a record of documented units, not a current inventory, so frames that have since left the author’s hands retain their entries, marked sold. Registry-derived generalizations are stated at **n = 7**, and every such generalization is made once, in §4.4, with other sections referring to it rather than restating it. Sony’s own 1999 service documentation supplies the second evidentiary pillar. Units documented by others are admitted behind an explicit scope wall (§4.5) and never enter registry counts.

Withholding. By the author’s standing ruling, no transaction figure of the author’s —

purchase, sale, or asking price, exact, banded, approximated, or recoverable by ratio — appears anywhere in this monograph; no date of any author transaction publishes at any resolution beyond the year 2026; the author’s sources publish only as three channel classes, a UAE electronics recycler, a Japan electronics recycler, and a US electronics recycler (the sealed BKAM-102 lot); sale particulars beyond the fact and configuration of a sale are withheld; buyers, sellers, and dealers are unnamed; and all such detail lives only in the author’s private records. Third-party figures and dated third-party observations are published exactly as observed. The remainder of the monograph obeys this ruling without further notice: a ledger row reading “confirmed — figure withheld” is documented, not unknown.

1.2 What the AMS-100 Is

The Sony AMS-100 — model plate text “Audio Monitor Speaker AMS-100,” made in Japan — is a 1RU rack-mountable audio confidence monitor built for professional video environments. It is not a loudspeaker in the hi-fi sense. It is a compact, rack-dwelling instrument whose job is to let an engineer *hear* the audio accompanying a video signal at the point in the plant where that signal is being monitored: a master control position, an edit suite, a quality-control bay, an outside-broadcast (OB) vehicle.

The unit’s defining architectural feature is its two-slot option-board system. The frame itself supplies stereo amplification and small internal speakers (2W + 2W), a headphone output, and a balanced monitor output; what it *listens to* is determined entirely by which BKAM-series option boards are installed in slots A and B — the three documented by Sony’s 1999 service literature are analog, AES/EBU digital, and embedded-SDI de-embedder [IM][MM]; a fourth, later board accepting HD-SDI is attested by a single photograph and is treated at §3.1 [social-media photo, unverified]. This modularity is the subject of Section 3 and is the key to understanding both why the AMS-100 existed and why surviving units vary so much in usefulness to modern owners.

1.3 Position in Sony’s Professional Monitoring Line

Sony’s professional video business of the era was anchored by its reference and picture monitors — the BVM (Broadcast Video Monitor) series at the reference-grade top, the PVM (Professional Video Monitor) series below it. These monitors displayed picture; they did not, in general, solve the problem of hearing program audio at the monitoring position, particularly for digital and embedded audio formats.

The AMS-100 filled that gap. Its 1RU form factor and rack ears fit it into the same equipment bays as terminal gear, and its physical proportions also suited stacking directly atop a CRT monitor — the configuration the author uses in practice (Section 5). The intended pairing with BVM/PVM-class displays is evident from the product’s positioning in dealer materials and from its I/O complement, which mirrors the signal formats found in broadcast plants rather than consumer or studio-music environments. [Confidence: high for the pairing use case as a general matter; the specific claim that Sony marketed it explicitly as a BVM/PVM companion has not been verified against period Sony literature.]

1.4 Predecessor and Lineage

Sony marketing copy links the AMS-100 to an earlier Sony monitor speaker, the AMS-3 — and the AMS-3 is no longer merely a name in that copy. **Four AMS-3 units have been physically documented** and are described in full in **Appendix E**: two Japanese domestic 100 V examples (serials 10685 and 11558) from contributor photographs [contributor photo, EXT-01], and two multi-voltage export examples (serials 20087 and 20144) handled and photographed by the author [author photo]. The two groups differ in model-plate part number, supply rating, corporate text, and serial band, and establish that the AMS-3 shipped in at least two market variants (§E.2). Across both variants it is: a 2RU, fixed-input, analog-only confidence monitor whose per-channel input-conditioning grammar — switched 600 Ω termination, switched input level, XLR loop-through — recognizably anticipates the vocabulary of the AMS-100's BKAM-101 analog board.

The lineage confidence flag nonetheless remains low, deliberately. [Confidence: low — the succession claim still derives solely from Sony marketing copy. Physical units confirm that the AMS-3 existed and what it was; they cannot confirm that Sony positioned the AMS-100 as its successor. The one documentary path the contributor intake offered — period paperwork accompanying the units — closed negative-by-unavailability, so the flag is movable only by a period cross-reference from another source: an AMS-3 document naming the AMS-100, or the reverse. Appendix E, §E.4–E.5, weighs the comparative evidence and the remaining paths.] If the connection holds, the AMS-100 represents the modular, digital-era successor to a simpler analog confidence monitor — a plausible trajectory given the industry's migration to AES/EBU and embedded SDI audio during the relevant period, and one the side-by-side comparison in Appendix E makes more concrete without making it sourced. Plausibility is not confirmation.

1.5 Era and Production Window

The window is fixed at one end. **The AMS-100 platform dates to no later than February 1999.** The anchor is documentary: the Installation Manual is a 1st Edition carrying Sony print code 1999-02, ©1999, covering the frame and all three BKAM boards as a single established product family [IM]; the Maintenance Manual's 1st Edition followed two months later (print code 1999-04) [MM]. A complete first-edition service documentation set is contemporaneous with product introduction, so the platform was shipping, or at the point of shipping, by early 1999.

Internal evidence corroborates the era from inside the hardware, and pushes activity at least into that year: **board production ran at least into 1999** [contributor photo, EXT-01 — a Crystal CS8412 receiver date-coded 9903 on a BKAM-102, oscillator codes 992005/992006 on a BKAM-103, and ©SONY99 firmware labels on the CPLDs of both boards; the component date readings are interpretations at moderate confidence, and the ©SONY99 marks establish a copyright floor only]. This is the project's first internal date evidence of any kind, and it is consistent with — indeed, interlocks with — the documentary $\leq$ February 1999 introduction [IM]. **The end of production remains unknown**, and the finding does not move it: a 1999 date floor inside boards from a platform introduced by February 1999 bounds nothing at the far

end.

That end is likely to remain unknown, because every external surface examined by this project is confirmed date-free. Every registry model plate inspected at close range carries no date code of any kind (§4.4); the plate of the externally documented frame 10047 (§4.5) is likewise date-free with text identical to the registry plates [contributor photo, EXT-01]; the three factory-sealed BKAM-102 cartons documented in §4.2 were inspected and carried no printed dates anywhere [author confirmed; the carton inspection is retrospective against discarded objects, corroborated for the visible faces by the author’s carton photograph, which is held in the project record and not published (Appendix C)]; and the four AMS-3 units of Appendix E carry date-free plates as well — across both of the market variants documented there — extending the observation back a product generation [contributor photo, EXT-01; author photo]. The uniform negative is itself a finding: **Sony’s external marking practice for this monitoring family — successor and predecessor candidate alike — omitted dates entirely.** The remaining evidence path is internal — board stamps or IC date codes — and the contributor board-pull above demonstrates both that the path is real and that it can yield without forcing this project’s own hardware: the registry units are working installations, and no board is pulled for dating purposes alone (§4.4). That conditional discipline stands.

Circumstantial evidence continues to bound the era from the design side. The option-board complement presupposes a plant built around serial digital video with embedded audio — the BKAM-103’s support for D2 composite serial digital alongside D1 component (§3.4) places the design in a period when D2 VTR islands were still live, squarely consistent with the documentary $\leq$ Feb 1999 date. And the documented units’ provenance — decommissioned-facility surplus surfacing in the mid-2020s (Sections 4, 7) — fits a retirement timeline for equipment installed roughly two to two-and-a-half decades earlier.

Two earlier low-confidence reports are formally retired here (resolved in §2.6): there was never a “230VAC 50Hz variant” — every unit carries a universal 100–240 V supply, with regional differentiation by cord set only [IM, plate-confirmed on hardware] — and the “PAL/NTSC switchability” report resolves to S102, the 525/625-line video-system select on the BKAM-103 board, which governs the de-embedder rather than the frame [IM; attribution author’s inference, confidence high].

1.6 Use Case: Why a Modular Confidence Monitor

In a broadcast or post-production plant, audio arrives at a monitoring position in whatever format the plant’s routing infrastructure carries: balanced analog on XLR, AES/EBU digital on 110 Ω XLR or 75 Ω BNC, or embedded within the SDI video stream itself. A fixed-input monitor speaker can serve only one of these worlds. The AMS-100’s two-slot design let a facility configure each unit for its rack’s actual signal environment — two analog boards for a legacy bay, an AES board and an SDI de-embedder for a digital master control position — and reconfigure as the plant evolved. Section 3 examines the boards individually; Section 5 shows how this plays out in a working installation today, including the practical consequence that a BKAM-103 is inert in an analog-only chain.

1.7 Why Document It Now

This monograph exists for three reasons, and its position on each is stated here once.

The documentation vacuum. The AMS-100 has no written history. The primary sources that exist — two Sony service manuals and a handful of dealer specification pages — are scattered and unindexed, and the Operation Manual has not been located at all (Appendix D). Everything in this work that is not a transcription of those sources is new.

Survival is uneven by channel, and provenance evidence is perishable. The position this monograph supports is: **rare in US-visible channels; documented Japan-domestic dealer and auction inventory as of July 2026** [dated sweep; point-in-time; no rate claim]. In US-visible channels a handful of transactions constitute the entire documented record (Section 7). The units that do surface there arrive through narrow decommissioning channels — notably Gulf-region facilities — and carry provenance evidence (tamper seals, as-surfaced configurations) that survives only until the units are handled onward (§4.4). The author’s 2026 acquisitions through a Japan electronics recycler (§4.1) are consistent with the Japan-domestic half of that position and are further dated observations, not a rate. No claim is made about how often AMS-100s surface anywhere.

Price discovery is nearly absent. As of this writing the author’s own transactions — seven purchases and three sales — are the best-documented AMS-100 transactions known, and the author’s sales are the first confirmed transactions in the record with the author on the selling side (§7.1–7.2). Against them stands a single active third-party US-visible listing, an ask, not a sale, at \$999.99. The BKAM-101 analog board — the member of the family with no confirmed per-board transaction anywhere — is now physically verified and working on the author’s bench at a census stated in §4.4 and §7.4; that census is hardware evidence, not market evidence, and changes nothing about the per-board transaction record.

Sections 4 and 7 address the documented units and the market respectively.

Sources for this section: Sony Installation Manual, 1st Ed., part 3-202-309-01, printed 1999-02 [IM]; Sony Maintenance Manual, 1st Ed., part 9-967-830-01, printed 1999-04 [MM]; a broadcast-equipment dealer’s specification pages (identity in the project record) [dealer — unverified where so flagged]; the author’s acquisition, sale, and intake records (private; intake records for frame 11353 and the fleet — see Appendix D); the author’s confirmation of the carton date inspection; the EXT-01 photo intake of July 2026 (frame 10047 plate; BKAM board internals; AMS-3 pair — see §4.5 and Appendix E) [contributor photo, EXT-01]; the author’s photographs and handling of the AMS-3 export pair, 2026 (Appendix E) [author; author photo]; the July 2026 literature sweep and the July 2026 market-listing sweep documented in the project facts file (the two are distinct: listings are not literature, and the primary-literature claim rests on the former). Predecessor claim: Sony marketing copy, unverified — see Appendix E. §1.5’s “covering the frame and all three BKAM boards” describes the Installation Manual’s own coverage and stands as written.

Section 2 — The Frame

Primary sources for this section: Sony AMS-100/BKAM-101/102/103 Installation Manual, 1st Edition, Sony part no. 3-202-309-01, ©1999 Sony Corporation Broadcasting & Professional Systems Company, printed Japan 1999-02 (textfiles.com STARIN manuals mirror) — cited [IM]; and the Sony AMS-100 Maintenance Manual, 1st Edition, part no. 9-967-830-01, ©1999, printed Japan 1999-04, 70 pp. (elektrotanya.com) — cited [MM]. Claims still resting on dealer listings only are marked [dealer — unverified]; every such claim is routed to the Operation Manual (§2.8). The Installation Manual print code resolves the introduction-date question raised in §1.5: the AMS-100 platform dates to no later than early 1999.

2.1 Overview

The AMS-100 frame is the constant in every configuration: a rack-width chassis containing the power supply, the stereo amplifier and internal speakers, the front-panel input-select and metering hardware, the headphone and monitor outputs, and two rear slots that accept the BKAM-series option boards. The frame itself has no program-audio inputs; without at least one option board installed, it can monitor nothing. Sony’s own documentation set reflects this frame-plus-boards architecture — the installation and maintenance manuals each cover the AMS-100 and all three BKAM boards together [IM][MM] (a fourth, later board, covered by neither manual, is attested only externally — §3.1).

2.2 Chassis and Dimensions

Manual dimension drawing gives a body width of 424 mm, an overall width of 482 mm (standard 19-inch rack width, i.e. including the rack-mount surfaces), and a depth of 350 mm [IM]. Height is not legible in the text extraction of the manual’s dimension drawing, but is hardware-confirmed: the frame measures at the nominal 1RU height (44 mm), verified on registry hardware [author, hardware-verified July 2026]. The chassis is made in Japan.

Serial numbering for the documented product family begins at 10001: the manual states applicability to “AMS-100 (SY) Serial No. 10001 and Higher,” with identical statements for all three BKAM boards [IM]; the maintenance manual’s schematics carry the same serial-applicability banner on every sheet [MM]. The (SY) suffix denotes the sales-region designation. This floor underlies the serial census in §4.4.

2.3 The Two-Slot Architecture

The rear panel carries two option slots. The manual states that the optional boards can be installed **in any slot** — there is no dedicated analog or digital position [IM]. Installation is a service-level task but a simple one: power off, remove a blank rear panel (two retainer screws), slide the board along internal guide rails, secure with the screws supplied with the board. The manual specifically instructs the installer to retain removed blank panels and to cover any unused slot [IM] — a detail with collector relevance, since frames that spent their service life fully populated may

have had their blanks discarded, and an unblanked empty slot is a sign of a later board pull. The registry’s experience of this tell is recorded at §4.4.

Internal configuration switches live on the boards, not the frame (BKAM-101 input-level and termination selects; BKAM-102 input select; BKAM-103 video-system and decoder switches — detailed in Sections 3 and 6). The frame’s own adjustable element accessible without disassembly is the input level meter calibration, performed from the front panel (Section 6).

2.4 Monitoring Path: Speakers, Headphones, Monitor Output

The frame provides:

Internal speakers and amplification — 2W + 2W stereo [dealer — unverified; not stated in either service document, which contain no published specifications table; Operation Manual item, §2.8].

Headphone output — -12 dBu at 8Ω [dealer — unverified, same basis; Operation Manual item, §2.8].

Monitor output — 2 channels. The nominal level is manual-verified: the maintenance manual’s circuit description states that the internal -10 dBs program bus is amplified to a **+4 dBm standard signal** for output on the REMOTE connector, and the corresponding service alignment specification is $+4.0 \pm 0.1$ dBm [MM]. The $+24$ dBm maximum and 600Ω balanced source figures remain dealer-only [dealer — unverified; Operation Manual item, §2.8]. The physical carriage of this output is not what one might assume: the monitor outputs are not on XLRs. They appear as balanced hot/cold/ground pin triplets (MONITOR OUT 1 and 2) on the rear **REMOTE** connector, a female D-sub 25-pin [IM]. A facility wanting to pass monitored audio downstream therefore had to break it out of the D-sub — consistent with the AMS-100’s role as an end-of-chain confidence monitor rather than a routing element.

Front panel — INPUT SELECT operation panel and a two-channel LED input level meter scaled $+3$ to -20 (0-reference calibration point) [IM]. The maintenance manual’s panel diagrams confirm the full button complement: SLOT A/B select, GP 1-4, CH-L 1-4, and CH-R 1-4 [MM] — the group- and channel-select controls whose role in embedded-audio monitoring is explained in §3.4. Full operational behavior is documented in the Operation Manual, which was supplied with the unit and remains a separate acquisition target for this project [IM lists it under Related Manuals].

2.5 Rear Panel and Remote Control

The REMOTE D-sub 25-pin connector carries, in addition to the two monitor output channels: remote channel-select inputs (L1/L2/R1/R2), group-select inputs (1/2), remote-enable inputs (1/2), and corresponding status (STS) outputs mirroring each select line [IM]. The maintenance manual adds electrical detail: all remote inputs and outputs are TTL-compatible, active low, and the paired SEL lines carry a 2-bit H/L code selecting among the four groups or channels [MM]. The full pin assignment is reproduced in Appendix B. The presence of remote select/status lines indicates the AMS-100 was designed for integration into monitored control surfaces or

tally/automation systems — an engineer could switch what the unit was listening to without touching its front panel. No evidence has yet been found of a dedicated Sony remote panel accessory for this connector [Confidence: absence of evidence only].

Board-dependent rear connectors (AUDIO IN 1/2 XLRs, DIGITAL AUDIO IN XLR/BNC, SDI IN and SDI OUT BNCs) belong to the option boards and are covered in Section 3. Note in passing that the manual’s connector table confirms the BKAM-103 provides an SDI **output** as well as an input — i.e., a loop-through, so the de-embedder could sit in-line in a video path rather than on a dead-end feed [IM].

Sony’s specified mating connectors: ITT Cannon XLR-3-12C (male) / XLR-3-11C (female) or equivalent for the XLR positions; 75Ω BNC with 5C-2V coaxial cable (Fujikura or equivalent, max. 200 m run) for the BNC positions [IM].

2.6 Power

The installation manual specifies: **AC 100-240 V $\pm$ 10%, 25 W consumption, 20 A inrush** [IM]. This is a universal-input supply.

No voltage variant. Earlier dealer-sourced research recorded a “230VAC 50Hz variant” of the AMS-100. The manual shows there is no separate voltage variant — every unit accepts 100-240V. Regional differentiation was by supplied power cord only: a US/Canada cord set, a UK cord set (DK-2401 UK), and a continental-Europe cord set (DK-2401 AE), each with a plug holder [IM]. The dealer listing that prompted the variant claim was likely describing a unit supplied with a European cord.

No video-standard switch on the frame. The single-source report of “PAL/NTSC switchability” is likewise explained by the manual. The setting in question is almost certainly S102 on the BKAM-103 SDI board, which selects between 525-line and 625-line video systems for the de-embedder (plus S103’s D1/D2 serial-format select) [IM; attribution of the dealer claim to this switch is the author’s inference, confidence high]. Section 3 covers these switches in detail.

2.7 Mounting: Rack vs. Stack

Rack. The AMS-100 mounts in a 19-inch standard rack, but not on simple rack ears: Sony requires the **RMM-30 rack mount rail kit** — sliding inner/outer rails with front and rear brackets, allowing the unit to be pulled forward out of the rack while cabled [IM]. The manual carries an explicit caution that rails other than the RMM-30 may lack sufficient strength and drop the unit. Practical consequences documented in the manual: leave enough rear cable slack for the unit to be pulled out; bolt the rack to the floor against tipping; recommended ventilation fan in the rack; and an operating range of 5-40 °C for all rack contents [IM]. For collectors, the RMM-30 requirement matters: ex-facility units frequently arrive without their rails (registry accompaniments, §4.1), and correct rails are a separate hunt.

Stack. Nothing in the installation manual addresses placing the unit atop a monitor; stacking on a CRT (the author’s configuration, Section 5) is a field practice, not a Sony-documented mounting mode. It is mechanically trivial — the unit is compact

and light relative to a broadcast CRT — but it forgoes the serviceability the rail system was designed for. [Author observation.]

2.8 The Documentation Ecosystem

The installation manual's Related Manuals section establishes a three-document set [IM]; two of the three are in hand:

1. **Installation Manual** — obtained July 2026 (textfiles.com STARIN mirror). Board installation, rack mounting, pin assignments, internal switches, level-meter adjustment. Service-personnel audience. 1st Edition, part no. 3-202-309-01, printed 1999-02, trilingual safety matter (English/German/French), J and E versions [IM].
2. **Operation Manual** — supplied with the unit; overview and system configuration examples. **Not located:** confirmed absent from the online manual aggregators at the July 2026 literature sweep, and the search is closed by author ruling — this monograph publishes without it. It is the one source that would have verified the amplifier, headphone, and residual monitor-output specifications resting on dealer listings (§2.4) and the operator-facing channel-selection procedure; those figures therefore stand permanently at [dealer — unverified].
3. **Maintenance Manual** — obtained July 2026 (elektrotanya.com): 1st Edition, part no. 9-967-830-01, ©1999, printed Japan 1999-04, 70 pages, covering service overview (error codes, circuit descriptions, maintenance modes), electrical alignment, spare parts, semiconductor pin assignments, block diagram, schematics, and board layouts for the frame and all three boards [MM]. It anchors Section 6. It does **not** contain a published specifications table, so it does not substitute for the Operation Manual on the remaining spec flags.

Sources: Sony AMS-100/BKAM-101/102/103 Installation Manual, 1st Ed., part no. 3-202-309-01, ©1999, printed Japan 1999-02 [IM]; Sony AMS-100 Maintenance Manual, 1st Ed., part no. 9-967-830-01, ©1999, printed Japan 1999-04 [MM]; dealer listings for the flagged residual specifications [dealer — unverified]; author measurement of registry hardware, July 2026 [author].

Section 3 — The Module System

Primary sources for this section: Sony AMS-100/BKAM-101/102/103 Installation Manual, 1st Edition, part no. 3-202-309-01, ©1999 [IM]; Sony AMS-100 Maintenance Manual, 1st Edition, part no. 9-967-830-01, ©1999, printed 1999-04 [MM]; dealer product listings (identity in the project record) for electrical I/O specifications not restated in either service document [dealer — unverified]; the author's seven-frame registry (Section 4; two members since sold, entries retained) for hardware and field-configuration evidence; one third-party social-media photograph (EXT-03, Appendix D) for the §3.1 fourth-board attestation [social-media photo, unverified].

3.1 Why a Modular Monitor

The AMS-100's defining design decision is that the frame carries no program-audio inputs of its own (Section 2.1). Every input arrives on an option board — the three documented by Sony's 1999 service literature are BKAM-101 (analog), BKAM-102 (AES/EBU), and BKAM-103 (embedded SD-SDI) — installed in either of two rear slots, in any combination, in any slot [IM]. A fourth, later board is attested by a single photograph and is treated at the end of this subsection.

In a late-1990s broadcast plant this was not a gimmick. Facilities of that era ran three audio transports simultaneously: legacy analog on balanced XLR, discrete AES/EBU digital, and audio embedded in the serial digital video stream. A monitor point in an analog-era edit bay, a digital audio machine room, and an SDI router bay needed three different input stages but the same speakers, meter, and headphone jack. Sony's answer was one frame SKU and, as documented in 1999, three board SKUs: engineering stocked one spare frame and a handful of boards rather than three distinct monitor models, and a bay could be migrated from analog to AES to embedded audio by swapping a card, not the unit. The two slots also allowed a single frame to bridge transitions — one analog board and one digital board in the same chassis, selectable from the front panel or the remote lines (Section 2.5).

The documentation set — a single installation manual and a single maintenance manual, each covering the frame and the three boards of the time, with a common serial floor of 10001 for frame and boards alike [IM][MM] — confirms the family was engineered and documented as one system rather than a frame with afterthought accessories. The maintenance manual adds the internal architecture that makes any-slot interchangeability work: every board delivers program audio to the frame as a -10 dBs standard two-channel signal on an identical 48-pin DIN slot connector, with per-board trimmers aligning each to that common level [MM].

A fourth board — HD-SDI [social-media photo, unverified]. Everything above describes the family as Sony documented it in 1999. One piece of evidence, outside every source the project has swept, indicates the family was later extended. A photograph posted publicly by a collector in Taiwan in August 2026 shows an AMS-100 rear panel carrying, in one slot, a module whose panel legend reads SDI IN over a BNC with the loop (buffer) symbol and an IN lock LED — the BKAM-103 layout of §3.4 — and, in the other, a module legended **HD SDI IN** (BNC), **INPUT MONITOR** (BNC), and **IN** (LED). The first module's panel label reads BKAM-10■, the final character obscured; a partial marking ending E0104 is visible near the second module, class unknown. The poster's caption describes the chain in use — an HDMI-to-HD-SDI converter feeding a BVM's HD-SDI input, the BVM's HD-SDI loop-out feeding the AMS-100 — with audio delivered through the AMS-100 from a 720p source [social-media caption, unverified]. All readings here are from the one image; the author has neither contacted the poster nor sought further material, by ruling, and no follow-up is planned.

What this supports, and at what confidence: (i) that an option board accepting HD-SDI was produced for this frame at some point after the 1999 first-edition manuals — the panel legend is explicit, and the statement in the seller's listing for frame 11353 (§4.3) that the 103 does not support HD-SDI now reads as a distinction between boards rather than a limit of the product [moderate, on one photograph]; (ii) that it carried

an INPUT MONITOR BNC rather than the 103's SDI OUT loop — a re-clocked or buffered monitor output in place of a passive loop, consistent with HD-SDI's tighter jitter budget, though this is inference from a panel legend [low]; (iii) its Sony model number, which **cannot be read** from the image and is not recoverable from any swept source — the project publishes it as BKAM-101 and assigns no digit. Whether the HD module is physically in the AMS-100 rather than an adjacent unit is supported by the frame's visible A B slot legend and a 101's 600 Ω switch legend in the same photograph, and by the caption, but is not independently confirmed [moderate]. Nothing is known of the board's silkscreen designation, assembly part number, silicon, introduction date, or market presence, and no claim about any of these is made. Because the board postdates both manuals, the spares-policy and internal-architecture statements of §6.1 and §7.5 — sourced to the MM's three assemblies — stand as written for the three documented boards and are silent on this one. The entry is carried at its tier permanently unless hardware or documentation surfaces from another source.

3.2 BKAM-101 — Analog Input Board

I/O: 2 channels, balanced analog audio on XLR-3-31 (in) with XLR-3-32 loop positions; the maintenance manual confirms the pairs (CN1/CN2, CN21/CN22) are wired as true loop-throughs — signal can be input to either connector of a pair [MM]. Nominal level selectable +4 dB or 0 dB — the selector values are printed on the schematic [MM] — with maximum +24 dBm and bridging input impedance of 10 k Ω resting on dealer listings [dealer — unverified]. 600 Ω termination is real and switchable, per below.

Internal identity and switches: the board silkscreen designation is **AL-42** [IM], author-verified on hardware alongside the Sony assembly part number **1-673-874-11** and PCB stock code S04-378T [author photo; consecutive with the 102's and 103's part numbers, §6.4]. The maintenance manual documents **four** user switches, two per channel [MM], and the physical board establishes that they split across two accessibility classes the manuals do not distinguish:

Switch	Function	Factory setting	Accessibility
S1 / S21	600 Ω input termination ON/OFF, ch 1 / ch 2	—	Module panel — legended ON/OFF beside each channel's XLR pair, operable in situ [author]
S2 / S22	Nominal input level +4 dB / 0 dB, ch 1 / ch 2	+4 dB [IM]	Internal — mid-board slides, board removal required [author]

The S2/S22 → channel 1/2 mapping is manual-confirmed [MM] and hardware-confirmed: the board silkscreen groups S2 with CH1/RV1 (“CH1 LEVEL”) and S22 with CH2/RV21 [author photo]. The termination toggles are **functionally verified**: exercised electrically on both channels of board 10505 in service, ON loading the

input down and OFF presenting the bridging level, exactly as a 600 Ω termination should behave [author confirmed — functional test]. The panel placement is operationally telling: Sony put the daisy-chain termination decision where an installer can reach it with the board racked, and left the level alignment inside — the same operator/setup split the 102’s panel-mounted input select expresses (§3.3). The per-channel alignment trimmers (RV1/RV21, 4.7 k Ω [author photo; MM gives the procedure only]) complete the board’s user-facing complement.

Role and scarcity: the BKAM-101 is the board that makes an AMS-100 a plug-and-play analog monitor, and it is the scarce one — a claim that is channel-scoped. Physically verified, working examples exist on the author’s hardware, the first documented anywhere, and every one of them came out of Japan-domestic supply (census and channel detail at §4.4). The panel of board 10505 confirmed the manuals’ description in full — two channels, each an XLR male/female true loop-through pair with a panel 600 Ω toggle, the panel bare of any label [author photo]; the later boards match at photo tier [author photo]. The market half of the thesis is untouched by those acquisitions: frames surfacing in **US-visible** channels arrive loaded with BKAM-102s and -103s, no US listing of a 101 has ever been observed, and no confirmed per-board BKAM-101 transaction is known anywhere — every verified example arrived inside a unit-level purchase, and the only loose-101 datum is a Japan dealer’s sold-out listing record (§7.4). The scarcity assessment rests on the intact US-visible absence and unfilled demand, channel-scoped [confidence: moderate for the US-visible void]; the Japan-domestic acquisitions verify the boards, not the channel’s depth, and support no rate claim [point-in-time, 2026]. Where a 101 cannot be reached, analog reaches the frame through an external A/D converter feeding a BKAM-102 — the registry’s original solution, since largely retired in favor of the real thing (Section 5).

3.3 BKAM-102 — AES/EBU Digital Input Board

I/O: 2 channels (one AES/EBU stream), accepted either as 110 Ω balanced on XLR-3-31 or as 75 Ω unbalanced on BNC [dealer for impedances; connector complement corroborated by the installation manual’s connector table [IM]; the 110 Ω figure is additionally corroborated at component level by the 110 Ω termination resistor (R102) on the XLR input path in the schematic [MM]]. Sony’s specified mating hardware applies: ITT Cannon XLR-3-12C, or 75 Ω BNC on 5C-2V coax to a maximum 200 m run [IM].

Internal identity and configuration: the board silkscreen designation is **DAC-37** [MM]. The installation manual records no user switch for this board; the maintenance manual documents one, and it is the board’s entire user configuration. **S101** selects between the XLR and BNC inputs of DIGITAL AUDIO IN [MM]; the factory alignment procedure has the technician set S101 to XLR and restore it afterward [MM]. A second switch position, S102, exists on the board layout but is not mounted [MM]. The board also carries a green LED (D102) behind the rear panel indicating AES/EBU input lock — a useful field diagnostic [MM].

The practical consequence for operators and collectors: a BKAM-102 that appears dead on one input may simply have S101 set to the other connector. **S101 is mounted on the module’s rear panel**, between the BNC and the XLR, operable with the board

racked [author photo; the registry’s S101 census (§5.6) was possible only because the switch is panel-visible]. It is an **operator-facing panel switch with no auto-detect and no frame front-panel indication** — so the dead-input trap is real but the diagnosis is a glance at the module panel (§5.6), and install-and-forget applies at the frame level, not the board panel.

Role: the workhorse. The board was obtainable in 2026 in a documented lot of three from a US electronics recycler (ledger at §7.2) — a lot since established as **factory-sealed new old stock in Sony cartons** (§4.2). In a plant, AES/EBU was the least perishable transport: it survived the analog decommissioning wave and predated full embedded-audio infrastructure, so 102-equipped frames stayed useful longest. The loose supply observed in 2026 therefore has at least two sources: harvested pulls (inferred from the UAE strip pattern, §3.6 — boards extracted before frames were surplussed went *somewhere*) and surviving unsold new stock (documented directly by the registry lot) [confidence: the pull mechanism is inference only; the NOS channel is documented, n = 1 lot]. Either way the 102 remains the practical basis for putting an AMS-100 back into service today. One registry frame, 12793, surfaced with its 102 still aboard — the single-frame counterexample to the pull reading, treated at §3.6; the registry’s 102 chains are at §4.2.

3.4 BKAM-103 — SDI De-embedder Board

I/O: serial digital video in and out on 75 Ω BNC — the manual’s connector table confirms both **SDI IN and SDI OUT**, i.e. a loop-through [IM] — with de-embedding of up to 16 audio channels from the SDI stream [dealer for the stated count; the 4-group $\times$ 4-channel architecture below is manual-derived and consistent with it]. The loop-through matters operationally: the board could be inserted in-line in a live video path, monitoring audio without terminating the feed, rather than requiring a dedicated dead-end output from a router. Like the BKAM-102, the board carries a green rear LED (D104) indicating SDI input lock [MM].

Internal identity and switches: board silkscreen **SDI-43** [IM], author-verified on hardware with Sony assembly part **1-673-876-11** and stock code S06-091A [author photo; corroborating the EXT-01 photo-tier readings, §4.5.1]. Documented switches [IM] — all three, unlike the 101’s termination pair and the 102’s S101, **genuinely internal**: mid-board, board removal required [author photo]:

Switch	Function	Factory setting	Accessibility
S101	Free-run adjustment	OFF	Internal (silkscreen FREE RUN) [author]
S102	Video system select, 525/625 lines	525	Internal [author]

Switch	Function	Factory setting	Accessibility
S103-1	Serial format: OFF = D1 (4:2:2 component), ON = D2 (4fsc composite)	OFF	Internal — four-position DIP block [author]
S103-2/-3/-4	Factory use only	—	Internal [author]

The board’s own silkscreen reads **S103** at the DIP block [author photo] — independent hardware corroboration of the §6.3 determination that the MM alignment text’s “S301-1” is a typo. The family’s switch complement thus splits cleanly into two classes: operator-facing decisions on module panels (101 termination, 102 input select) and setup/alignment switches inside (101 level, all of the 103’s) — a distinction the manuals never draw but the hardware does [author].

The maintenance manual’s alignment section confirms the function of S101 (set ON, with a test point and trimmers, to adjust the serial decoder’s free-run frequency; returned OFF after alignment) [MM]. Two switches repay comment. **S102** is, with high confidence, the origin of the dealer-listing claim that the AMS-100 is “PAL/NTSC switchable” (resolved in Section 2.6): the video-standard selection exists, but it lives on this board and governs the de-embedder, not the frame [IM; attribution is the author’s inference]. **S103-1** is a period marker: support for D2 composite serial digital alongside D1/SMPTE 259M component places the board’s design squarely in a late-1990s plant where D2 VTR islands were still live — consistent with the $\leq$ February 1999 introduction date established by the manual print code.

Channel selection: the maintenance manual’s circuit description and panel documentation establish the mechanism. The de-embedder extracts one embedded audio **group** of four channels from the SDI stream; a two-channel pair is then selected from that group for the monitor path [MM]. On the front panel, the **GP 1-4** buttons select the group and the **CH-L 1-4 / CH-R 1-4** buttons select which channel of the group feeds the left and right monitor legs respectively [MM] — four groups of four channels covering the full 16-channel embedded complement, matching SMPTE embedded-audio group structure. The REMOTE connector’s group- and channel-select lines mirror the same choices via a 2-bit code per selector [MM]. The operator-facing procedure wording would rest on the Operation Manual, which is not located (§2.8); the engineering mechanism is established.

Role: the board most likely to still be aboard a decommissioned frame — in the channels where audio boards are harvested (the logic at §3.6; the as-surfaced census at §4.4). In an analog-only collector installation, the BKAM-103 is inert: with no SDI source, it de-embeds nothing, and it cannot be repurposed. The registry’s 103s are present-but-unused; their chains are at §4.2 and their installation status at Section 5.

3.5 Installation Mechanics

Board installation is covered in Section 2.3 and summarized here for completeness: power off, remove the slot’s blank panel (two retainer screws), slide the board on

internal guide rails, secure with the supplied screws; boards fit either slot; retain and reinstall blank panels over unused slots [IM]. The blank-panel instruction doubles as a provenance tell — a frame with an open, unblanked slot almost certainly had a board pulled after leaving Sony’s intended configuration discipline (Section 2.3).

3.6 Configurations and Scarcity

The evidence base is small — seven registry frames, one externally documented frame, and dated listing observations — and every reading below is flagged accordingly [confidence: low-to-moderate throughout]. The registry’s as-surfaced, serial, and board censuses are stated once, at §4.4; the market record by component is at §7.4. This subsection carries the configuration logic those two sections cite: what the design anticipated, what decommissioned frames arrive with and why, and which configuration cannot be built at will.

What the design anticipated. 2× BKAM-102 for a dual-AES monitor point; 103 + 102 for an SDI bay with a discrete-AES fallback; 2× 101 or 101 + 102 for analog and transitional rooms. The registry has run four of these layouts at one time or another (§4.1), and two are documented surviving in the wild as-surfaced: 103 + 102 on the externally documented frame 10047 (§4.5.1) and 2× 101 on frame 12886 (§4.3).

What decommissioned frames arrive with — the pull logic. A frame leaves service when a bay or a plant is retired, and what it carries out the door is decided by what its boards are worth elsewhere. The BKAM-101 serves any analog point and the BKAM-102 any AES point — AES being the transport that outlived the others (§3.3) — so both are worth pulling for reuse before the frame is surplussed. The BKAM-103 is useful only with an SDI plant and has the least standalone resale identity of the three, so it is the board left aboard. The result is the **harvest signature**: a single-board frame carrying a 103 with the other slot blanked. Both UAE-sourced frames arrived exactly so, and a Yahoo Auctions JP frame observed in the July 2026 sweep showed the same arrangement — 103 in slot A, slot B blank [auction listing, date unrecorded] — so the signature appears in Japan-domestic auction channels too [confidence: moderate for the pattern’s existence; low for any estimate of its prevalence]. Harvested pulls are also the inferred bulk source of the loose 102 supply (§3.3).

Channel-dependence. The signature is a real and recurring pattern, present in at least two channels, and not the fate of every frame. Japan-domestic supply, through the recycler channel the author used, delivered something else: a frame with its 102 still aboard (12793), a complete 101 + 103 frame (11353), frames stripped *to the analog board* (101-only), and one not stripped at all (2× 101, 12886); the externally documented Australian frame 10047 likewise arrived complete (§4.5.1). Every complete survivor in the record stands outside the UAE channel, and only one 103 surfaced from the Japan recycler channel [counts at §4.4; observation, no inference]. The correct statement is therefore that **strip-to-the-103 is a UAE-channel signature, not a universal fate, and which board is left aboard varies by channel** [confidence: moderate for the UAE-channel reading; low for any per-channel generalization]. Where 101s were left in frames, the pull economics that should have harvested them did not operate — whether because an installation retired as a unit, because the board had no local reuse, or for reasons the record cannot see, is not

determinable.

The configuration that cannot be built at will. Anything involving a BKAM-101 — a statement that requires its qualifications stated in full, because the author has built several. A loose 101 *was* purchasable from a Japan dealer before the July 2026 sweep found its listing sold out; 101-equipped frames were purchasable in Japan in 2026, and the author bought them, verified the boards, and runs them (§4.1) — so 101-based configurations are demonstrably buildable *when Japan-domestic supply happens to be reachable*. What remains true: they are unsourceable on demand through any visible per-board market — zero confirmed per-board transactions, one sold-out listing record, and every verified example inside a unit-level purchase constitute the entire supply record (§7.4); the author’s acquisitions are dated events through channels of unknown depth and generalize nothing about repeatability [point-in-time; the “lucky grab” caution stands for every one of them]. The practical modern substitute — external A/D feeding a BKAM-102 — remains the path available to anyone who cannot reach that supply (Section 5).

Section 4 — Documented Units

Primary sources: the author’s photo intakes and fleet photographs [author photo]; the author’s statements on configurations, board moves, and sales [author statement]; the author’s private transaction records (authoritative for figures and dates — none published); the author’s carton photograph [author photo; held in the project record, not published — Appendix C]; the archived seller listing photographs for frame 11353 [seller listing photo]; the Installation Manual [IM] for the serial floor and blank-panel discipline; and, for §4.5, the EXT-01 photo intake of July 2026 [contributor photo, EXT-01] and owner statements relayed by the author for EXT-02 [report tier].

4.1 The Registry

The registry comprises **seven AMS-100 frames** acquired by the author in 2026 through two supply channels: a UAE electronics recycler (two frames, one package) and a Japan electronics recycler (five frames). Three frames have since been sold. **Sold frames keep their full entries, marked sold:** the registry is a record of documented units, not a current inventory, and a board whose chain ends at a sale is still a documented board. Every generalization drawn from the registry carries its caveat — **n = 7** — and externally documented units (§4.5) stand behind a scope wall and do not move that n.

Frames are identified throughout this monograph **by frame serial**. The record numbers used during drafting (R-1 ... R-7) are mapped to frame serials once, in Appendix D, and appear nowhere else in the text; the author’s private record identifiers are not published. Transaction figures, dates, and counterparties are withheld from publication in every form by standing ruling (§1.1); each acquisition and sale recorded here is nonetheless a confirmed event documented in the author’s private record.

Registry of record (2026). Board identifiers are five-digit Sony PCB serials where read, seven-digit barcode-class labels (§4.4) for the two UAE-package BKAM-103s,

and the bare type designation where a board's serial is unread.

Frame	Channel	As-surfaced	Current	Status
10010	UAE electronics recycler (package with 11341)	103 (0105241) + blank	101 (10505) + 101	held
11341	UAE electronics recycler (package with 10010)	103 (0128521) + blank	101 + 103 (10631, by deduction)	held
11353	Japan electronics recycler	101 (10505) = A + 103 (10631) = B	102 (10099) = A + blank — as sold	sold
11878	Japan electronics recycler	101 = A + blank	102 (10154) = A + blank — as sold	sold
12793	Japan electronics recycler	102 + blank	103 (0105241) = A + 101 = B	held
12886	Japan electronics recycler	101 + 101 (both original — corroborated by the board census, §4.2)	101 = A + 102 = B	held
13048	Japan electronics recycler	101 = A + blank	103 (0128521) = A + 102 = B — as sold	sold

Supporting facts, by frame. **Seals** — 11341: one (C094183), broken by the author at a later board swap, fragment discarded; 10010: two at arrival (one over the 103, broken at the author's rebuild, serial unrecorded, discarded; C094182 across a non-opening seam, in place as last checked); the five Japan-sourced frames: none at any point [author; 11353 also seller listing photo]. **Plate anomalies** — 10010 carries a misapplied barcode-class label (0105235) over its plate text; none elsewhere. **Condition** — all seven working at last handling [author; 11353 also seller's test]. **Accompaniments** — no RMM-30 rails, paperwork, or Operation Manual with any frame [author]. **Evidence tier** — 10010, 11341, 12793, 11353: dedicated close-range photo intakes; 11878: the fleet photographs [author photo, held unpublished — Appendix C] and author statements only, no dedicated intake; 12886 and 13048: the fleet photographs and author statements for their as-surfaced state, with close-range rear photographs of their final configurations [author photo — plates C-40, C-35].

4.2 Board Provenance

Boards move between frames in this registry, so board histories are recorded as **chains** rather than locations. Twelve boards have passed through the author’s hands: nine arrived in frames, three came as a loose lot, and none has been purchased loose otherwise.

Board	Type	Identifier	Chain	Disposition
103	BKAM-103	label 0105241	UAE recycler → 10010 (as-surfaced, sealed) → author stock → 12793	installed, 12793
103	BKAM-103	label 0128521	UAE recycler → 11341 (as-surfaced, under seal C094183) → 13048 (seal broken) → sold with 13048	sold
103	BKAM-103	s/n 10631	Japan recycler → 11353 slot B → author stock → 11341	installed, 11341 [identity by deduction — see §4.3]
102	BKAM-102	s/n 10099	carton S01-0010099 (lot) → 10010 → 11353 → sold with 11353	sold
102	BKAM-102	s/n 10154	carton S01-0010154 (lot) → 10010 or 11341 → 11878 → sold with 11878	sold
102	BKAM-102	s/n 10083 (by elimination)	carton S01-0010083 (lot) → 10010 or 11341 → author stock → 12886 or 13048	installed; PCB label never read (serial by elimination stands — closed)

Board	Type	Identifier	Chain	Disposition
102	BKAM-102	unread	Japan recycler → 12793 (as-surfaced) → 12886 or 13048	installed; one of this pair sold with 13048
101	BKAM-101	s/n 10505	Japan recycler → 11353 slot A → 10010	installed, 10010 — working
101	BKAM-101	unread	Japan recycler → 11878 (as-surfaced) → author stock → registry frame	installed — working
101	BKAM-101	unread	Japan recycler → 13048 (as-surfaced) → author stock → registry frame	installed — working
101 ×2	BKAM-101	unread	Japan recycler → 12886 (as-surfaced, both original)	one installed, 12886 slot A; one moved to another registry frame — working

Four of the five BKAM-101s carry unread serials, and their individual routing is not stated: two arrived in 12886, one each in 11878 and 13048, and the four are now distributed one apiece to 10010, 11341, 12793, and 12886, alongside 10505 in 10010's other slot [author statement]. The net position: **five BKAM-101s, all working** [author — functional]; **three BKAM-103s**, two installed and one sold with frame 13048; **four BKAM-102s**, two installed — in 12886, and in 13048 at its sale — and two sold with frames 11353 and 11878.

The board census corroborates the registry's weakest reading. Frame 12886's as-surfaced configuration — BKAM-101 in *both* slots — was recorded from a fleet photograph taken at distance and an author statement, the thinnest evidence tier in the registry, and the frame's current configuration no longer shows it (§4.3). The arithmetic requires it independently: five BKAM-101s are documented, three of them traceable to other frames — 10505 out of 11353, one out of 11878, one out of 13048 — leaving two that can have originated nowhere but 12886. Had that frame arrived with a single 101, the census would stand at four and a documented working board would be unaccounted for. A distance photograph and a board count, arrived at by wholly independent routes, agree [confidence: moderate-to-high, upgraded from the photograph alone]. The same arithmetic closes the other side of the ledger: the two BKAM-102s that ended in 12886 and 13048 are 12793's as-surfaced board and lot board 10083, which is therefore **not loose**. Which of the two went to which frame is

unstated, and neither PCB was read before installation.

The BKAM-102 lot. The three lot boards came in a single transaction from a US electronics recycler (particulars withheld). They arrived **factory-sealed in Sony cartons** [author photo, held unpublished — Appendix C]: part number P-96604680-L on all three, a yellow “B-AVL” stock sticker with handwritten marks of unidentified meaning, and per-carton serials S01-0010154 / S01-0010099 / S01-0010083 (as-read; trailing check character uncertain on two). The cartons were discarded after a date inspection that returned a confirmed negative — no printed date on any face or flap [author confirmed; retrospective against discarded objects, the author’s carton photograph corroborating the visible faces]. The lot is the only confirmed per-board transaction known for any BKAM module, and it is a **new-old-stock** transaction — direct evidence that a NOS supply channel existed alongside decommission pulls (§7.4).

Carton serials are board serials — demonstrated. The carton identifiers parse as five-digit serials with an “S01” line prefix and zero padding. Lot board 10099, removed from frame 10010 and photographed in passing, carries a five-digit “SERIAL NO. 10099” label on its PCB component side — an exact match to carton S01-0010099 [author photo; the board’s lot provenance is fixed by 10010’s rebuild history]. Board 10154 was read by the author at its later move, a second exact match [author statement]. The carton numbers and the PCB “SERIAL NO.” labels are therefore **one BKAM-102 serial line in two locations** — on the PCB always, on the carton for sealed stock — and BKAM-102s leave the factory panel-bare with their serial on the PCB face [demonstrated on two boards; the third, 10083, is matched by elimination only — it was installed into a frame without its PCB label being read, one of the two candidate frames has since been sold, and the mapping therefore closes permanently at two of three]. The check-character uncertainty on the carton readings concerned the trailing character only and is irrelevant to the five-digit match.

4.3 Notes on Individual Frames

What the tables cannot carry. Entries are limited to facts with evidentiary weight beyond the frame itself.

10010 — The lowest AMS-100 serial known to this project, nine above the documented floor of 10001 [IM]. Arrived in the same package as 11341 with one BKAM-103 and **two** tamper seals — one over the 103 and one across a chassis seam that opens nothing — the latter establishing that the sealing practice attached to units, not to boards (§4.4). Its plate carries a white barcode label reading **0105235**, stuck sloppily over the product-name and AC-rating lines: format-identical to the seven-digit class on the UAE 103 panels, it is a board-class label misapplied to a frame [author ruling], and it cannot have come from this frame’s own 103, whose label 0105241 is intact with no second-label ghost. Reading, flagged not ruled: 0105235 is the trace of a fourth BKAM-103 processed by the same handler, six below 0105241 in the class numbering [author inference; low-to-moderate]. The label is recorded in the author’s photographs and retained in place. This frame has had the registry’s deepest intervention: the as-surfaced 103 removed (seal broken) and two lot 102s installed at the author’s rebuild; one 102 later swapped for 101 (10505); the other subsequently swapped for a second 101.

11341 — Arrived with one BKAM-103 (label 0128521) under a holographic tamper seal, “WARRANTY PROTECTION / DO NOT REMOVE,” serial **C094183**, bridging the chassis corner and the board panel edge [author photo, held unpublished — Appendix C]. C094183 is **exactly sequential** with 10010’s surviving seal C094182 — the finding that resolves the seal class to a single third-party issuer applying seals in series (§4.4). While it stood, its placement across the board-to-chassis seam certified that 0128521 had not left the frame between the seal’s application and the author’s ownership. The author broke the seal at a later board swap to move 0128521 into frame 13048, and discarded the fragment [author statement]; the sequential-pair finding rests on the recorded serials and is unaffected. The frame’s current 103 is **s/n 10631, ex-11353, out of storage — by deduction**: 0128521 is in 13048 and 10631 was the only other loose 103 the author held [author deduction; slot letters not stated].

12793 — The first Japan-recycler frame, and the registry’s only frame to surface carrying the liquid board rather than the frame-ballast 103; it carried no seal at any point [author] — the first Japan-channel datum against the UAE strip pattern (§4.4). No purchased board was added to it. The author installed the 103 pulled from 10010 (label 0105241) alongside the as-surfaced 102; the 102 has since moved to another frame and a BKAM-101 stands in slot B, so the one frame that surfaced carrying a digital input board now runs none [author statement].

11353 — The registry’s first complete-as-surfaced frame: BKAM-101 (10505) in slot A, BKAM-103 (10631) in slot B, **confirmed at physical intake against the seller’s listing photographs** — the monograph’s one calibration point for how listing-photo configuration claims are weighted, and it verified cleanly. The slot assignment itself stands at listing-photo tier permanently: both boards were removed on intake day before the arrangement was re-photographed (plates C-22/C-23 are the record). The 101 was the first physically verified BKAM-101 anywhere and, once installed in 10010, the first documented working example, with the 600 Ω termination exercised electrically on both channels [author confirmed — functional test]. Seller’s condition notes at listing: working; volume-potentiometer crackle; meter uncalibrated [seller listing]. The 103’s CPLD firmware (IC102 V1.10 / IC105 V1.00) diverges from EXT-01’s example (V1.00 / V1.00) — the record’s first evidence of firmware revision within a board type, carried with its discrepancy flag in §6.4. Reconfigured on intake day to lot 102 (10099) + blank; sold in that configuration, S101 at XLR, working [author statement].

11878 and 13048 — Both arrived from the Japan recycler as **101-only** frames, slot B blanked: single-board frames stripped to the *analog* board, the inverse of the UAE harvest signature (§4.4). Both 101s work [author — functional]; neither serial was read, and neither board’s 600 Ω as-found positions were recorded before power-on. After the fleet photograph the author removed both 101s into 10010/11341. 11878 received lot 102 (10154) and was sold. 13048 — **the highest AMS-100 serial known, 3,047 above the floor** — received 103 label 0128521 in slot A (the author’s placement, excluded from the as-surfaced slot census). The Sony blank first fitted to slot B was later replaced by a BKAM-102, and the frame was **sold in that two-board configuration**, the 103 untested in it throughout [author statement; author photo — plate C-35].

12886 — Arrived from the same channel with **BKAM-101 in both slots, both original to the frame**: the registry’s second complete-as-surfaced frame, the record’s

third (with EXT-01's 10047), and the first 2× 101 frame documented anywhere — the pure analog layout §3.6 lists among the design's anticipated configurations, surviving in the wild. That reading was first recorded at the registry's weakest tier — a fleet photograph at distance plus an author statement — and the author's own "believed original" qualifier is now discharged by the board census, which requires two 101s to have originated here (§4.2). Both boards worked; serials unread.

The frame has since been reconfigured: one 101 moved out to another registry frame and a BKAM-102 took slot B, leaving **BKAM-101 (A) + BKAM-102 (B)** [author photo — plate C-40]. It is the registry's only frame documented in the 101 + 102 layout, and it holds the only installed BKAM-102 remaining in the author's hands. The as-surfaced 2× 101 finding is unaffected: §4.4's censuses count frames as they surfaced, not as they stand.

4.4 Cross-Registry Findings

As-surfaced configuration is channel-dependent. The UAE pair arrived 103-only — resale-worthy audio boards pulled, the SDI de-embedder left aboard, the harvest signature §3.6 describes. The Japan recycler channel delivered something else with every frame: 12793 arrived 102-only; 11353 arrived complete (101 + 103); 11878, 13048, and 12886 arrived 101-only, 101-only, and 2× 101 complete. Complete frames thus stand at **n = 3 (11353, 12886, and EXT-01's 10047), all outside the UAE channel**, and only one 103 surfaced from the Japan channel at n = 5. The strip-to-the-103 pattern is a **UAE-channel signature, not a universal fate** [confidence: moderate; n as stated, one channel sampled at n = 2].

Serials and the production floor. All seven frame serials are read — 10010, 11341, 11353, 11878, 12793, 12886, 13048 — and with the external frame 10047 the census runs 10010 < 10047 < 11341 < 11353 < 11878 < 12793 < 12886 < 13048, every one above the documented floor of 10001 [IM]. The registry brackets that floor from 9 to 3,047 above it, and on the untested single-sequential-series assumption **13048 sets a production-volume floor of roughly 3,050 frames** [author inference; confidence low-to-moderate — the sequential assumption is untested, and the figure is a floor, not an estimate of output]. Frame 10010 reads as approximately the tenth frame produced under the same assumption.

The seal system: third-party, single-issuer, UAE-channel. Three seals are known, all on the UAE pair; no Japan-sourced frame carried one at any point. C094182 (10010, non-opening seam, surviving) and C094183 (11341, board seam, since broken) are **exactly consecutive serials on two frames purchased in one package** — the discriminating evidence for one issuer applying seals in series [confidence: moderate-to-high]. The seals are generic commodity foils with no Sony branding and a serial format matching no Sony scheme on the hardware; the author's ruling, adopted as the leading reading, is that they are **supply-chain artifacts of the UAE channel** — a facility, dealer, or handler sealing its fleet — not Sony factory or service marks [author ruling July 2026; issuer unknown]. 10010's second seal, across a seam that opens nothing, shows the practice sealed *units* at multiple seams, deterrent-fashion. The cross-channel test — does any non-UAE frame carry such a seal? — has been run six times and passed six: 12793 [author], 11353 [author

intake + dealer listing photo, photographed surfaces], 11878 / 12886 / 13048 [author statement], and EXT-01's 10047 [contributor photo, all 22 images, both slot seams after a board pull — a photo-tier negative, permanent]. The falsification path stands: a matching seal on a frame from an unrelated channel would reopen the question. One datum is lost permanently — the serial of 10010's discarded first seal, plausibly adjacent to the surviving pair, a hypothesis the record can state but never test.

Three board-labelling systems. (1) **Five-digit “SERIAL NO.” labels on the PCB** — the true Sony board serial line, in the 10001+ band the Installation Manual declares for all four products of the family [IM]. Author-verified on all three documented board types (10505 on a 101, component side; 10099 and 10154 on 102s, component side; 10631 on a 103, solder side) and at contributor tier on EXT-01's boards (10016, 102; 10130, 103); tied to Sony's carton serialization by the §4.2 demonstration [confidence: high]. Placement is consistent within type at $n = 2$ each. (2) **Seven-digit white barcode labels** (0105235 / 0105241 / 0128521) — an asset, inventory, or lot class of open identity, **not** serials. Every example in the record traces to the single UAE package: the 103s from the Japan recycler and EXT-01, and all four Japan-recycler 101s, are panel-bare [author photo; contributor photo]. The class reads as labelling applied by one downstream handler in the UAE channel, consistent with the seal finding; 0105235's misapplication to a frame plate is the record's one direct evidence of label handling in the supply chain [author; low-weight caveat]. (3) **Paper labels on the CN101 backplane headers** — 030281 (101), 951756 (102), 0703BJ (103) [author photo, July 2026]. Six characters, one alphanumeric; candidate reading, flagged not ruled: a connector-vendor lot or date mark on the connector body rather than a Sony board identifier. Nothing in hand discriminates.

Plates carry no dates. All seven registry plates are format-identical except serials — “AC100-240V~ 50/60Hz 0.5A,” label part 3-703-002-02, origin mark “JAPAN” — and none of the four examined at close range carries a date code or region suffix; the same seven readings corroborate the manual's universal-supply statement at the hardware level — every plate is marked for 100-240 V operation, with no voltage variant among them (§2.6); the three August plates show none at the fleet photograph's distance, which is not a close-range negative [author declined close-up plate shots]. With EXT-01's 10047, three cartons, and the four AMS-3 plates of Appendix E — covering both of that product's documented market variants — the uniform negative spans **eight AMS-100 plates**: Sony's external marking practice for this family omitted dates entirely (§1.5).

Slot arrangements. As-surfaced, the 103-in-A convention held 3/3 (11341, 12793, 10047) and 11353 inverted it (101 = A, 103 = B) [seller listing photo]; every frame that arrived with a single 101 carried it in slot A (11353, 11878, 13048), and 12886 filled both. All counts in this subsection are as-surfaced and are unaffected by the later reconfigurations of 12793, 12886, and 13048 (§4.3). Author placements — the August swaps and 0128521 into 13048's slot A — are excluded from the as-surfaced count. Any board fits any slot [IM]; the conventions are installers' habits, not constraints.

Blank panels. Every single-board frame, in every channel sampled, arrived with its empty slot covered by a Sony blank plate (§2.3). Disposition of the plates removed at the early rebuilds is unrecorded [minor].

Open questions from the hardware phase. The registry boards from the first three frames were never inspected board-side — the author declined to pull working boards for it [ruling July 2026] — so the five-digit status of 103 labels 0105241/0128521 and of 12793's as-surfaced 102 is unknown; the question has since been largely overtaken by the author-tier census above. The region fields in the author's private records for the first three frames match no consistent scheme and are read as entry artifacts; all three chassis sweeps were negative, so reconciliation rests on seller documentation [flagged, unresolved]. The bounds on this section's record are stated in Appendix F.

4.5 Externally Documented Units

Scope wall. The units below are **not registry members**. They carry no record numbers, have received no author hardware verification, and contribute nothing to registry-derived generalizations. Evidence from them arrives at two tiers — **[contributor photo, EXT-*nn*]**, read directly from photographs, and **[contributor report, EXT-*nn*]**, relayed statements — and contributor credit lives once, in Appendix D, per contributor preference. External units are the monograph's only instrument for testing whether registry findings are properties of the AMS-100 population or artifacts of one collector's purchases; each entry states what it corroborates, extends, or complicates. No external unit has, or will have, a Section 7 ledger row.

4.5.1 EXT-01 — frame 10047, Australia

One AMS-100, frame serial **10047**, held in Australia; configuration as photographed **BKAM-103 in slot A, BKAM-102 in slot B** [contributor photo, EXT-01; plates C-10-C-18] — the first two-board AMS-100 photographed anywhere, and the second-lowest serial known (46 above the floor). Its plate is textually identical to the registry plates with no date code or market suffix [plate C-11].

The contributor removed and photographed both boards, component and solder sides, producing the first open-board photographs of any BKAM module and three findings. **First**, both boards carry Sony-format five-digit "SERIAL NO." labels on the PCB — 10130 (103, solder side), 10016 (102, component side) — the evidence that first established the five-digit class as the true serial line (§4.4). **Second**, the project's first internal date evidence: the 102's CS8412 receiver date-coded 9903, the 103's oscillators 992005/992006 [1999-family; moderate], and ©SONY99 firmware labels on all three CPLDs [copyright floor only] — board production ran at least into 1999 (§1.5). **Third**, board identity new to the record at the time — assembly parts 1-673-876-11 (103 / SDI-43) and 1-673-875-11 (102 / DAC-37), stock codes S94-510A and S92-042A — corroborating the Maintenance Manual's board designations at photo tier (§6.4), where this unit's firmware complement (V1.00 / V1.00) is also flagged against 11353's.

Provenance artifacts: a "B-STOCK" sticker on the 102's solder side — a dealer or refurbisher grading mark, placing a resale channel somewhere in the board's history [interpretation moderate] — and a captive BNC→RCA adapter on the 102's panel BNC, independent evidence of the consumer-coax field practice also documented on the registry installation (§5.2). **No seal** of any class appears in any image, including both slot seams after the board pull — a photo-tier negative, permanent, and one of the six cross-channel passes of §4.4.

Recorded as states, final: acquisition date, venue, channel, and price **not disclosed**; working condition **not tested or reported**; Australia is a **custody datum only**, not a supply-channel finding. The same contributor's two AMS-3 units are documented in Appendix E.

4.5.2 EXT-02 — Los Angeles, USA (status entry)

One AMS-100 **reported** to carry a BKAM-101, held in Los Angeles; non-working; offered for sale, then withdrawn when the owner tested it; the owner retains it and declines to sell in any condition [owner statements relayed by the author, July 2026 — report tier; no photographs]. The fault is externally indeterminate — frame-side (power supply, AU-265 mute path) and board-side (the 101's protection-IC links [MM]) are both plausible, and nothing here should be read as “a failed BKAM-101.” Two items remain open: the withdrawn asking price (which would enter the Section 7 ledger as an ask-withdrawn, never a sale) and the failure symptoms, interpretable against the §6.2 blink-code table. A photo pack would make this the first documented BKAM-101 in third-party private hands and the record's only documented failed example. The entry converts to a full record on photo receipt and stays unverified until then.

Section 5 — Integration in Practice

Primary sources for this section: the Sony Installation Manual [IM] and Maintenance Manual [MM] as cited; the registry (Section 4) for the as-built installation; author field practice, flagged [author] where a claim rests on installation experience rather than Sony documentation. This section describes one working 2026 collector installation as it stands; it is descriptive, not prescriptive.

5.1 The Installation Problem

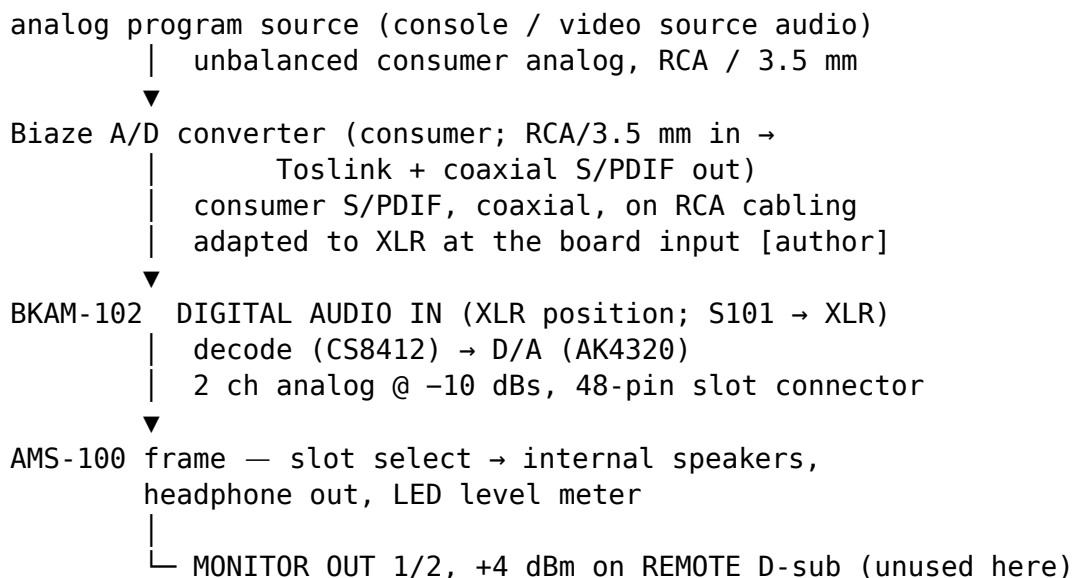
The registry installation is an analog-video monitoring suite: broadcast CRTs (one PVM, two BVMs) fed by analog sources, with an AMS-100 at each display providing confidence audio. The natural board for this job is the BKAM-101 analog input board, and the installation now runs one at **every** monitor point (§4.1). That is a recent state, reached in two steps: until the Japan-recycler acquisitions no BKAM-101 could be sourced and every point ran the substitute described below; the 101s then arrived, only ever through unit-level purchases (§4.2, §7.4); and the last 102-fed point converted to analog when its BKAM-102 was moved to another frame [author statement].

The substitute — which served every point in this installation at one time or another, and the last of them until 2026 — is the one identified in §3.6: an external analog-to-digital converter feeding the obtainable BKAM-102 AES/EBU board. This inverts the board's original role. In a late-1990s plant the 102 monitored an already-digital transport; here digital is manufactured solely so the AMS-100 can hear an analog source. But it is the configuration the surviving parts supply supported, it remains the path for anyone without access to that supply (§3.6), and it exercises the frame

exactly as designed: the frame neither knows nor cares which board type delivers its −10 dBs program feed (§3.1) [MM for the internal standard].

5.2 The Signal Path

The chain below describes the installation’s last 102-fed point, frame 12793 (§5.3), as it ran until that frame converted to analog. **No monitor point in this installation is 102-fed today.** The chain is recorded in full regardless: it is the documented form of the substitute path §3.6 identifies for anyone who cannot reach BKAM-101 supply, and every reading and caution below was taken on working hardware while it ran.



Three properties of this chain are worth stating explicitly.

The digital hop is consumer S/PDIF, not AES/EBU proper. Source and monitor transducers are analog; the digital hop exists solely to cross the BKAM-102’s input stage, and it is a consumer-format hop. The converter is a **Biaze consumer analog-to-digital audio converter, model HW-US-Z1**, listed as “Analog to Digital Audio Converter, RCA to Optical Converter RCA/3.5mm to Toslink and Coaxial Adapter for PS5 PS4 HDTV DVD Soundbar” [author]. It has no AES/EBU output: what reaches the BKAM-102 is **consumer S/PDIF from the converter’s coaxial output, carried on RCA cabling and adapted to the board’s XLR DIGITAL AUDIO IN** [author]. Sample rate: the manufacturer’s specification supports 32, 44.1, and 48 kHz, 48 kHz maximum [manufacturer listing, via author] — bounding the chain’s operating rate to three values, all within the CS8412/AK4320 decode chain’s capability, the 48 kHz case matching the board’s service alignment reference [MM]. The as-running rate is not instrumented; the item is closed at specification level [author]. That the chain works rests on receiver tolerance: the CS8412 accepts both professional (AES-3) and consumer (S/PDIF) frame formats, and its input sensitivity accommodates a consumer-level signal into the board’s 110 Ω-terminated balanced input at these cable lengths [author, from the part’s published characteristics]. On paper the arrangement is out of spec against both AES-3 and the S/PDIF interconnect standard — level, impedance, and balance are all mismatched — and it is documented here as verified-

working field practice, not as a recommendation [author]. The same chain ran at all three monitor points through July 2026 [author, bench record].

Input-side levels are fixed by the converter — at consumer alignment. With a BKAM-101, nominal level is selected on-board (+4/0 dB, S2/S22 [MM]); in this chain, analog headroom is set by the Biaze’s fixed consumer input stage (the nominal –10 dBV world), and the AMS-100’s meter reads the digital-domain result against its –20 dBFS reference with a 102 aboard [IM]. A consumer device carries no professional calibration (+4 dBu = –20 dBFS), so meter indications at this point are relative rather than aligned — reinforcing the meter’s confidence-indicator role (§5.6) [author].

The monitor outputs are unused. The +4 dBm MONITOR OUT pairs exist only as pin triplets on the REMOTE D-sub (§2.4) [IM][MM]; in an end-of-chain confidence-monitoring role there is nothing downstream, and the installation leaves the D-sub unconnected. No remote select/tally integration is used; all source selection is front-panel.

The practice is not unique to this installation. The externally documented AMS-100 of §4.5.1 (EXT-01, frame 10047 — non-registry) arrived with a captive screw-on BNC→RCA adapter left in place on its BKAM-102’s BNC DIGITAL AUDIO IN — physical evidence that that board, too, was at some point fed consumer coax on RCA cabling [contributor photo, EXT-01; plate C-16]. Read as an independent second instance of the consumer-coax-into-BKAM-102 field practice documented above [interpretation, mod]. The corroboration is strictly photo-tier and says nothing about this installation’s chain, which stands on its own bench verification; nor is the adapter’s upstream equipment or signal format known — only the connector adaptation is in evidence. What it suggests is that pressing the 102’s tolerant receiver into consumer-coax service is a solution more than one owner has reached independently.

5.3 The Monitor Points

Current configurations are those of the registry table (§4.1); board identities and chains are at §4.2.

Frame 10010 → PVM-20L5. Two BKAM-101s (10505 + one unread). The point takes analog direct on both slots, selected from the front-panel SLOT A/B buttons [MM for the button complement]. The 600 Ω termination and analog pass-through were exercised on 10505 at its installation [author confirmed — functional test]; the second board works [author — functional]. Before the 101s arrived this was the installation’s dual-AES point — two 102s, each fed by its own converter, genuinely two sources rather than one source split [author, bench record].

Frame 11341 → BVM-14 Master. One BKAM-101 (unread) and one BKAM-103 (10631, by deduction — §4.3); slot letters not stated. The point takes analog direct on its 101 and carries an inert 103 (§5.4).

Frame 12793 → BVM-20E1U. BKAM-103 (0105241) + BKAM-101, the 101 active. Until 2026 the second slot held a BKAM-102 and this was the installation’s last 102-fed point — the point the §5.2 chain describes; that board moved to another frame and the position now takes analog direct [author statement]. The monitor pairing

recorded earlier in this project (a BVM-A14) is superseded by the author’s later statement. As it ran, the two BVM positions were fed jointly: the same program feed, split in the digital domain — a single converter serving both, its output passively divided with an XLR splitter immediately ahead of the two DIGITAL AUDIO IN connectors [author]. A note on that split, since it is field practice several steps from by-the-book transport: a passive Y places both boards’ 110 Ω input terminations (R102 [MM]) in parallel, roughly halving the load the converter’s consumer-level S/PDIF output drives — compounding the format and level mismatches already documented in §5.2. At confidence-monitor cable lengths, with the CS8412’s tolerant receiver, the arrangement was verified working; a distribution amplifier or transformer splitter would be the by-the-book alternative if runs lengthen or lock proves marginal [author]. Because the audio was identical at both positions, the assignment was treated as operationally immaterial; the physical pairings are as recorded here [author, July 2026]. With 12793 now 101-fed as well, no converter is documented in service at any monitor point; the disposition of the converters is not recorded (Appendix F).

Each monitoring AMS-100 sits stacked atop its CRT and monitors via its internal speakers, with the headphone jack available for isolation checks [author]. **Frame 12886** — BKAM-101 in slot A and BKAM-102 in slot B, the registry’s only installed 102 (§4.3) — holds no documented monitor point; fourth point, bench unit, or reserve is not stated [author; open]. Its 102 is not documented as fed by anything, and the installation as described here does not use it.

The cabling of the 101-fed points (10010, 11341, 12793 — which sources feed which slot, slot-selection practice), the disposition of the converters, and the splitter state at 12793 at the time of its conversion are undocumented [author] and stay so by ruling (Appendix F).

5.4 Why the BKAM-103 Is Inert Here

Frames 11341 and 12793 each carry a BKAM-103 alongside a working input board (§4.1; board histories at §4.2). In this installation both are inert, and it is worth being precise about why, since “inert” is a statement about the plant, not the board:

- The BKAM-103 accepts exactly one input class: serial digital video with embedded audio, on 75 Ω BNC [IM]. An analog video installation has no SDI anywhere; there is no adapter path, because the missing element is the transport itself, not a connector.
- The board cannot be repurposed as a generic digital-audio input. Its decode chain (serial decoder CXB1342R, group de-embedding CXD8280AQ [MM]) begins at an SDI deserializer; discrete AES/EBU is the BKAM-102’s job.
- Selecting a 103’s slot from the front panel with no SDI lock present yields silence; the D104 rear LED (§5.6) will be dark, correctly reporting no input [MM for the LED].

The boards nevertheless stay installed. They occupy slots that would otherwise need blank panels (§2.3), they keep both boards in documented, registry-tracked homes (§4.2), and they keep an upgrade path open: introducing any SDI source to the suite would make both boards live without a single purchase. Present-but-unused is the correct description, not surplus [author].

5.5 Mounting and Cabling Practice

Mounting. The monitoring units are stacked atop their CRTs — the field practice noted in §2.7, which owns the rack-vs-stack treatment. Two practical observations from this installation [author]: the AMS-100’s 350 mm depth and modest weight sit comfortably on a broadcast CRT’s cabinet, but placement should respect the CRT’s top ventilation louvers — broadcast monitors shed substantial heat upward, and the AMS-100’s own 5–40 °C rating [IM] assumes it is not parked over an exhaust path. Position the unit toward the cabinet front or on standoffs as the specific CRT dictates. Second, the stacked unit forgoes the RMM-30 rail system’s pull-out serviceability entirely; board swaps mean lifting the unit down, which at this size is a non-event.

Cabling. The digital interconnect is the consumer-S/PDIF-to-XLR adaptation described in §5.2 — RCA cabling from the converter’s coaxial output, adapted to the board’s XLR-3-31 input (Sony’s specified mating connector being ITT Cannon XLR-3-12C male [IM]) [author]. The board-end electrical facts stand regardless of source format: the XLR path is 110 Ω -terminated (R102 [MM], corroborating the dealer 110 Ω figure). A by-the-book alternative — a true AES/EBU source on purpose-rated 110 Ω balanced cable — remains available should lock ever prove marginal [author]. The BNC path (75 Ω , 5C-2V, ≤ 200 m [IM]) is unused in this installation, the one installed BKAM-102 being switched to XLR (§5.6). Cabling at the 101-fed points: §5.3, open.

5.6 Gotchas and Field Diagnostics

The Maintenance Manual turned several previously invisible board behaviors into checkable facts; this installation has verified the first of them directly.

S101 input select — the “dead input” trap. Each BKAM-102 accepts AES on *either* XLR or BNC, chosen by the toggle S101, which is mounted **on the module’s rear panel**, between the BNC and XLR, readable and operable without pulling the board [author photo; §3.3]. There is no auto-detect and no *frame* front-panel indication: a board switched to BNC presents a perfectly dead XLR, indistinguishable at the connector from a failed board. The diagnosis is therefore one glance: **read the toggle at the module panel** before suspecting the board, the cable, or the source. The registry’s one installed BKAM-102 now sits in frame 12886 (§4.3), which has no documented monitor point, and its S101 position is not recorded. The last 102 in monitoring service — 12793’s, until its removal — was set to XLR as last checked [author].

Lock LEDs as first-line diagnostics. Both digital boards carry a green rear-panel LED reporting input lock: D102 on the BKAM-102 (AES/EBU lock) and D104 on the BKAM-103 (SDI lock) [MM]. In practice this gives a two-step fault split visible without opening anything: LED lit + no audio $\rightarrow$ look downstream (slot selection, frame, muting); LED dark $\rightarrow$ look upstream (source, cable, S101 position). The LEDs face the cable field, so on a stacked unit they are directly inspectable; in a rack they justify the RMM-30 pull-out access.

Power-on mute. The frame implements a power-on mute circuit [MM]; a moment of silence at switch-on is designed behavior, not a fault. Relevant to a suite that is powered up per-session rather than running continuously [author].

Meter trust and warm-up. The front-panel meter reads the -10 dBs internal bus and is calibratable from the front panel; Sony specifies a 10-minute warm-up before calibration [IM]. For session use the meter is a confidence indicator, not a measurement instrument, and the -20 dBFS reference (with a 102 aboard) is the anchor for interpreting it (§5.2).

Channel-select buttons with a BKAM-102 aboard. The GP and CH-L/CH-R buttons exist for the BKAM-103's 4-group $\times$ 4-channel de-embedding (§3.4) [MM]. Their behavior, if any, when the selected slot holds a two-channel BKAM-102 is not documented in either service manual, and the Operation Manual is not located (§2.8). In this installation the question is cosmetic — the 102 delivers its two channels regardless — and it is not investigated by experiment (Appendix F).

Sources: Installation Manual [IM]; Maintenance Manual [MM]; registry and board chains (Section 4); author bench and installation records [author]; EXT-01 photo intake, plate C-16 [contributor photo, EXT-01].

Section 6 — Service & Internals

Primary sources for this section: Sony AMS-100 Maintenance Manual, 1st Edition, part no. 9-967-830-01, ©1999 Sony Corporation Broadcasting & Professional Systems Company, printed Japan 1999-04, 70 pp. [MM] — the anchor document; and the Installation Manual [IM] for installation mechanics and the front-panel meter calibration procedure. §6.4 additionally carries board-level identity data read from the EXT-01 open-board photographs, tagged [contributor photo, EXT-01] (second-party, non-registry — see §4.5 for the scope wall), and from the author's own boards [author photo]. All other part numbers, IC designations, alignment values, and mode/error tables below are transcribed from the maintenance manual's service overview, alignment, spare-parts, and schematic sections. Division of labor with Section 5: §5.6 owns field-level diagnostics (lock LEDs, the S101 input-select trap, power-on mute as observed behavior); this section owns the service-level layer beneath them. Cross-references are given rather than duplicating that material.

6.1 Service Overview and the Serviceability Design

The maintenance manual covers the frame and the three option boards it documents — BKAM-101, BKAM-102, and BKAM-103 — in one 70-page document, organized as: service overview (error codes, circuit descriptions, maintenance modes), electrical alignment, spare parts, semiconductor pin assignments, block diagram, schematics, and board layouts [MM]. It contains **no published specifications table** — the residual dealer-only spec flags (§2.4, §3.2–3.3) therefore survive it and still route to the Operation Manual. **Scope of this section.** A fourth, later board accepting HD-SDI is attested by a single photograph and postdates both manuals (§3.1) [social-media photo, unverified]; nothing about its internals, service, or spares treatment is known, and every [MM]-sourced statement in this section — including the spares policy below — is made for the three documented boards only.

Internal architecture. The serviceable design rests on the standardized internal interface already introduced in §3.1: every documented option board delivers program audio to the frame as a two-channel analog signal at the **−10 dBs internal standard**, on an identical 48-pin DIN slot connector [MM]. On the frame side, slot A/B selection is performed by analog switching at IC202/IC203; the selected program is amplified by IC204/IC205 to the **+4 dBm standard** for the MONITOR OUT triplets on the REMOTE D-sub (§2.4), while the headphone amplifier and level meter are fed at the −10 dBs bus level [MM]. A power-on mute circuit suppresses output at switch-on (§5.6 for the field-facing note) [MM]. The consequence for service work is that each of the three documented board types is interchangeable at one defined electrical plane, and frame-side faults can be isolated from board-side faults at that plane; whether the HD-SDI board of §3.1 meets the same plane is undocumented.

Board installation is a service task in Sony’s taxonomy but a mechanically simple one — power off, blank panel off, board onto guide rails, two screws — covered in §2.3 and §3.5 [IM]; it is not repeated here.

Spares policy — boards are not field-repair items. The manual’s parts documentation carries a significant policy line: the option boards are **not spare-stocked as bare PCBs**. The prescribed remedy for a failed board assembly is replacement of the whole product — AL-42 → BKAM-101, DAC-37 → BKAM-102, SDI-43 → BKAM-103 [MM]. Sony’s own service model thus treated board and product as one unit of exchange, which has a direct 2026 echo: the loose BKAM-102s circulating on the secondary market (§3.3, §4.2) *are* the service spare, and a dead BKAM-101 has no repair path through Sony parts at all — one more contributor to that board’s effective scarcity (§3.2) [MM for the policy; the market reading is the author’s]. The policy is documented for the three 1999 assemblies only; the manual predates the HD-SDI board (§3.1) and is silent on it.

6.2 Maintenance Modes and Error Codes

The frame’s HD64180ZR CPU (§6.4) gives the AMS-100 a small but genuine self-test and configuration layer, entered by holding front-panel button combinations while switching **POWER** on [MM]:

Buttons held at power-on	Mode
GP4 + CH-R3	SLOT status output (STS-A/STS-B, REMOTE pins 14/15) set to level type
GP4 + CH-R4	SLOT status output (STS-A/STS-B, REMOTE pins 14/15) set to pulse type (500 ms low)
GP3 + CH-R1	Button / LED test mode
—	RAM check mode (listed in the manual alongside the above) [MM]

The STS modes configure the REMOTE connector’s **SLOT status output** (§2.5, Appendix B): the manual states the type selection for the status signal of the SLOT A/B

select button, output on pins 14/15 — either a continuous level mirroring the select state, or a 500 ms active-low pulse on change [MM §1-5-1]. Whether the remaining STS lines (the channel- and group-select status pairs) share the configured type is **unstated in the MM** — open, resolvable by the Operation Manual or a bench probe. This remains the one piece of persistent configuration that lives in the frame rather than on a board switch, and it is invisible without either this manual or a logic probe on the D-sub — worth recording during hardware intake if remote integration is ever attempted.

Error blink codes. At self-test failure the frame reports faults by blinking front-panel indicators at a 200 ms rate; the manual maps each pattern to a specific frame IC [MM]:

Blinking indicator(s)	Fault	Implicated IC
GP1 + SLOT	Board hardware error	—
GP2	ROM checksum	IC115
GP3	RAM read/write	IC114
GP4	Flash read/write	IC113
CH-L1	PIO 0	IC116
CH-L2	PIO 1	IC117

Documentation erratum (author determination). The table above transcribes the manual’s §1-3 error table as printed — including its parenthetical IC attributions, which contradict the manual’s own parts list, semiconductor data, and §1-4 circuit description. Those sections establish **IC113 = M27C512 EPROM** (part 8-759-587-30, labeled “M27C512-AU265-IC113-V1.00” — the firmware ROM), **IC114 = SST29EE512 flash** (§1-4 verbatim: “backup data in the flash memory of IC114”), and **IC115 = LC361000AMLL SRAM**. The blink-to-function mapping (GP2 = ROM checksum, GP3 = RAM read/write, GP4 = flash read/write) is not itself in question; the printed IC numbers appear rotated by one position. Reconciled against the manual’s internal evidence, the implicated devices are **GP2 → IC113 (ROM), GP3 → IC115 (SRAM), GP4 → IC114 (flash)** [MM for all passages; the reconciliation is the author’s, exactly parallel to the §6.3 S103-1 determination]. The stake is practical: a technician chasing a GP2 blink to IC115 as printed would probe the SRAM rather than the ROM.

For a collector, this table converts a blinking, otherwise-dead frame from a mystery into a component-level diagnosis at power-on, with no test equipment. It complements rather than overlaps the field-level fault split in §5.6 (lock LEDs, S101): the blink codes report the *frame’s* health; the rear LEDs report the *input’s*.

6.3 Electrical Alignment

The manual’s alignment section specifies test equipment including the Sony/Tektronix SG-5010 audio generator, AM-700 audio measurement set, TSG-422 and WFM601M for the video side, and BKPF-201/105A interface boards for generating the SDI test signal the BKAM-103 procedures require [MM]. The procedures relevant to this project:

Frame — monitor output. Service specification for the MONITOR OUT level is **+4.0 ± 0.1 dBm** [MM] — the figure that verified the nominal monitor-output level in §2.4.

Frame — level meter. The meter is calibrated from the front panel without disassembly: hold the slot button for 3 seconds to enter calibration [IM]. Reference input is **+4 dBm** with a BKAM-101 aboard or **−20 dBFS** with a BKAM-102, and Sony specifies a **10-minute warm-up** before calibration [IM]. (§5.2 and §5.6 cover what these references mean for the as-built installation.)

BKAM-101 (AL-42). Per-channel trimmers RV1 (“CH1 LEVEL”) and RV21 align each channel to the −10 dBs internal standard, measured as **347 ± 3 mV at CN101 pins 19/35** [MM]. These are the trimmers that make any-slot interchangeability real (§3.1): each board is set to the common plane, not to a particular frame.

BKAM-102 (DAC-37). Alignment reference: **1 kHz at −20.0 dBFS, 48 kHz sample rate, emphasis off**, aligned for **+4.0 ± 0.1 dBm** at the monitor output [MM]. The procedure has the technician set the S101 input toggle to XLR for the test and restore it afterward (§3.3) — the service-manual origin of the “dead input” trap documented in §5.6.

BKAM-103 (SDI-43) — free-run alignment. With S101 set ON, the serial decoder’s free-run frequency is adjusted at test point TP102: **RV102 → 26.9 ± 0.1 MHz** for D1 component operation, **RV101 → 14.2 ± 0.1 MHz** for D2 composite. After alignment S101 is restored to OFF and the serial-format switch returned to D1 [MM].

Documentation erratum (author determination; hardware-corroborated). The BKAM-103 alignment text refers to the D1/D2 serial-format switch as “S301-1”. The board layout contains no S301; the switch is **S103-1** (§3.4 table). The alignment-text designation is a typo, determined by checking the reference against the manual’s own board layout [MM for both passages; the reconciliation is the author’s] — and corroborated on hardware: the physical board’s silkscreen at the DIP block reads **S103** [author photo; BKAM-103 s/n 10631, received in frame 11353 (§4.2)]. Anyone following the procedure verbatim on the bench should substitute accordingly.

Safety. The manual specifies a leakage-current test with a **≤ 3.5 mA** limit [MM] — standard practice for line-powered service work and applicable after any PSU-adjacent repair.

6.4 Boards, Semiconductors, and Parts

Frame board set. The frame comprises five boards [MM]:

Board	Function	Notable components
AU-265	Main board: CPU, analog switching, power amplifier, monitor-output drivers	CPU HD64180ZR (8-bit); slot select IC202/203; monitor drivers IC204/205; speaker amp IC201 = TA8200AH/TA8216H (dual designation as recorded in the parts data)
MT-136	Level meter	BA682A drivers; 2× DC-3H7GWA LED arrays
HP-97	Headphone amplifier	NJM4556AD; RV1 level pot; mute logic
SW-10	Front-panel illuminated buttons	Dynamic scan drive
VR-257	Volume control	20 kΩ / 20 kΩ pot

The power supply is a switching regulator, and the two internal speakers carry Sony part number **1-529-353-11** [MM] — the reference for any future driver replacement.

Option-board silicon. The digital boards’ decode chains, cited functionally in Sections 3 and 5, are: BKAM-102 — CS8412 AES/EBU receiver into an AK4320 D/A; BKAM-103 — CXB1342R serial decoder, CXD8280AQ time-base expansion performing the group de-embed, into the same AK4320 D/A [MM]. The shared D/A part is one more expression of the common –10 dBs output plane.

Board identity from hardware. The EXT-01 intake produced the first open-board photographs of any BKAM module [contributor photo, EXT-01]; the author’s own boards put **author-verified hardware of all three documented board types** on the record [author photo] — including the BKAM-101, whose internals had never been documented at any tier. Corroborated against [MM]: the **AL-42**, **DAC-37**, and **SDI-43** board designations, all read directly from PCBs in the author’s hands, and the **CS8412** AES/EBU receiver physically sighted (package CS8412-CS, on both the EXT-01 board and the author’s board). Board provenance and placements are given in §4.2; the identity record stands:

Board	Sony assembly part no.	PCB stock code	CPLD complement (Sony firmware label)	Example / tier
BKAM-101 (AL-42)	1-673-874-11	S04-378T	none — analog board; op-amp complement JRC 4560DD ×2 (IC1/IC2), A5201A ×2 (IC3/IC4)	s/n 10505 [author] — first identity record for this board type anywhere
BKAM-102 (DAC-37)	1-673-875-11	S92-042A	EPM7064S (IC101) — fw V1.10 ©SONY99	EXT-01 s/n 10016 [contributor photo]
BKAM-102 (DAC-37)	1-673-875-11	S96-547A	EPM7032S (IC101) — fw V1.20 ©SONY99	s/n 10099 [author]
BKAM-103 (SDI-43)	1-673-876-11	S94-510A	EPM7096 (IC105) + EPM7064S (IC102) — fw V1.00 ©SONY99 (both, as recorded)	EXT-01 s/n 10130 [contributor photo — intake record; see flag]
BKAM-103 (SDI-43)	1-673-876-11	S06-091A	EPM7096 (IC105) fw V1.00 + EPM7064S (IC102) fw V1.10 , both ©SONY99	s/n 10631 [author]

The **assembly part numbers run consecutively — 874 / 875 / 876, one per board type** — independent part-numbering evidence that the three documented boards were released as a single family, corroborating §3.1’s one-system reading from a new direction [author-verified across all three]. The author’s 102 additionally yielded a 12.288 MHz reference crystal (X101) and a TP103 “RXOUT” test point; the author’s 103 confirmed the V-24M576 24.576 MHz oscillator [EXT-01’s part, now author-verified]. The 103’s large QFPs — the CXB1342R/CXD8280AQ candidates — remain **[MM]-tier only on all examples**: not legible in the EXT-01 photographs, not legible at the author’s intake resolution (Appendix F).

Two cross-example discrepancies — flagged, not reconciled. With two examples

per digital board type, the record shows divergence, and in both cases one side of the comparison is the weaker tier:

1. **BKAM-102 CPLD part and firmware.** EXT-01's 102 was recorded as **EPM7064S, fw V1.10**; the author's board carries **EPM7032S, fw V1.20** [author photo — legible]. If both readings hold, the 102 shipped with at least two CPLD parts and two firmware states across production — a substitution of a smaller-macrocell part in the same IC101 position, with a firmware step. The EXT-01 reading is re-inspectable against its embedded source image (plate C-15, Appendix C). Also divergent between the two examples: the PCB stock codes (S92-042A vs S96-547A), consistent with distinct board revisions and a caution against presuming stock codes stable across production.
2. **BKAM-103 per-CPLD firmware.** EXT-01's 103 was recorded as **V1.00 on both CPLDs**; the author's reads **V1.10 on IC102** (EPM7064S), **V1.00 on IC105** [author photo — legible]. If both hold, the 103 shipped in at least two firmware states — the record's first evidence of revision activity within a board type. The asymmetry: the EXT-01 reading stands at [contributor photo — intake record] tier, and its source image is not embedded (plate C-13 is caption-only, Appendix C), so this comparison, unlike the first, cannot be checked against a published image.

Neither divergence is adopted as a production-history finding until its weak side is re-verified; both are recorded so that neither reading silently overwrites the other [flag-don't-reconcile, per standing project rule]; no re-verification is scheduled — both readings stand as recorded (Appendix F). The **©SONY99 marks remain firmware copyright years: a floor, not a manufacture date.** The manufacture-dating reading proper — board production ran at least into 1999 [EXT-01 component codes] — is carried in §1.5 and unchanged; **two 2000-family date-code candidates from the author's boards (an IC code on the 101, an oscillator code on the 103) are held in the facts file, deliberately unadopted — no code-format ruling or board sweep is scheduled** [author photo; not adopted — §1.5, Appendix F]. Scope caveats: $n = 1$ (101, author) / $n = 2$ (102, 103 — one author + one contributor each); part numbers verified stable across two examples per digital board type; stock codes demonstrated *not* stable.

Protection — IC links. Each board carries IC-link fuses (PS-numbered), with part numbers distinct by board family [MM]:

Board	IC links	Sony part no.
BKAM-101	PS101, PS102	1-533-984-11
BKAM-102, BKAM-103	PS101, PS102, PS103	1-533-282-21

These are the first suspects on a board that is entirely dead (no lock LED, no output on a known-good input): a blown IC link is a component-level repair well below the whole-product exchange Sony's spares policy otherwise prescribes (§6.1) [MM for the parts; the triage note is the author's].

Coverage for the appendices. The manual additionally contains full exploded views, parts lists for frame and boards, complete schematics, and board layouts [MM]. These

are the source base for Appendix A's spec tables (manual-derived figures, dealer-only figures flagged) and are not reproduced in body text.

Sources: Sony AMS-100 Maintenance Manual, 1st Ed., part no. 9-967-830-01, ©1999, printed Japan 1999-04 [MM]; Sony AMS-100 Installation Manual, part no. 3-202-309-01, 1st Ed., ©1999 [IM]; EXT-01 open-board photographs [contributor photo, EXT-01]; author board photographs [author photo].

Section 7 — Rarity & Market

7.1 The State of the Evidence

No editorial or community literature on the AMS-100 has been found — no reviews, no substantive forum threads, no collector documentation [literature sweep, July 2026; standing periodic re-sweep only]. This monograph is, as far as can be determined, the primary literature, and nothing below affects that claim: listings are not literature.

The market record splits by channel. **In US-visible channels** it is nearly silent: of the eleven frame sale events ledgered in §7.2, ten have the author as a party — seven purchases and three sales, all in 2026 — with a single third-party event (a US hi-fi reseller, ≈ 2017) beside them. **In Japan-domestic channels**, a dated market-listing sweep (July 2026) documented dealer and auction inventory the US-visible record gave no hint of (§7.6), and the author's five acquisitions through a Japan electronics recycler are a second observation in the same direction. Both are point-in-time findings for 2026; they support no claim about how often such inventory surfaces, and none is made in this monograph [documented listings, dated; rate claims barred by author ruling]. Every generalization in this section carries the n-of-few caveat [confidence: low-to-moderate throughout, flagged per claim].

Asking prices and confirmed sold prices are strictly distinguished throughout: an ask is a number someone typed; a sold is money that moved. A third state, the **listing record** — a dealer page marked sold out at a posted price, with no transaction confirmable — is neither, and is ledgered separately. The author's own transactions are published as undated events with figures withheld (§1.1).

7.2 The Ledger

Confirmed solds

Date	Item	Configuration / detail	Price	Channel
2026	AMS-100 frame 11341 — author purchase	1 × BKAM-103 only	confirmed — figure withheld	UAE electronics recycler

Date	Item	Configuration / detail	Price	Channel
2026	AMS-100 frame 10010 — author purchase	1 × BKAM-103 only	confirmed — figure withheld	UAE electronics recycler
2026	BKAM-102 ×3, loose lot — sealed NOS in Sony cartons (\$4.2) — author purchase	—	confirmed — figure withheld	US electronics recycler
2026	AMS-100 frame 12793 — author purchase	1 × BKAM-102 only	confirmed — figure withheld	Japan electronics recycler
2026	AMS-100 frame 11353 — author purchase	1 × BKAM-101 + 1 × BKAM-103 — listed configuration confirmed at physical intake (\$4.3)	confirmed — figure withheld	Japan electronics recycler
2026	AMS-100 frame 11878 — author purchase	1 × BKAM-101 only	confirmed — figure withheld	Japan electronics recycler
2026	AMS-100 frame 12886 — author purchase	2 × BKAM-101	confirmed — figure withheld	Japan electronics recycler
2026	AMS-100 frame 13048 — author purchase	1 × BKAM-101 only	confirmed — figure withheld	Japan electronics recycler
2026	AMS-100 frame 11353 — author as seller	1 × BKAM-102 (s/n 10099) + blank; working	confirmed — figure withheld	particulars withheld
2026	AMS-100 frame 11878 — author as seller	1 × BKAM-102 (s/n 10154) + blank; working	confirmed — figure withheld	particulars withheld

Date	Item	Configuration / detail	Price	Channel
2026	AMS-100 frame 13048 — author as seller	1 × BKAM-103 (label 0128521) + 1 × BKAM-102; frame working, the 103 untested in this frame (\$4.3)	confirmed — figure withheld	particulars withheld
≈ Nov 2017	AMS-100, one unit	cosmetic 7/Good; configuration unknown	price unknown	US hi-fi reseller

The US hi-fi entry is footnote-grade: its sold-out listing proves one unit moved through a US hi-fi channel around 2017 (photo timestamps), but with no price it contributes an existence datum only. The frame 11353 purchase is the ledger’s only row to carry a listing-tier configuration claim through to hardware verification; it came back clean (\$4.3).

Listing records (neither asks nor solds)

Observed	Item	Posted price	Status
July 2026 sweep (listing and sale dates unknown)	Loose BKAM-101 — “Analog Audio Module”	¥13,200 incl. tax	page marked SOLD OUT , under-negotiation badge — [dealer listing, sold out at listed price; not a confirmed transaction] — a Tokyo used-broadcast dealer

This single row is the only BKAM-101 price datum of any kind known to this project (\$7.4). The posted price is a dealer’s exact, publishable figure; whether a sale completed, when, and at what final price are unknown.

Asks (not solds)

Observed	Item	Price	Status
July 2026	AMS-100, one eBay listing (US-visible), third party	\$999.99	ask as observed July 2026; not re-observed

Adjacent unverified inventory, recorded rather than ledgered: an Asian export dealer claims five AMS-100s with BKAM-103s in stock, “pictures for reference, ship randomly” [dealer claim — unverified]; a Korean eBay seller lists an AMS-100 and two loose BKAM-103 modules [listing, July 2026 sweep]. Neither carries transaction weight; both bear on the \$7.4 component picture.

Stale data (excluded from all columns)

Prior research surfaced eBay figures of roughly **\$125-127**: undated catalog data of unknown vintage and unknown ask/sold status, excluded from the ledger rather than averaged into it, and retained here only so a future researcher encountering them knows they were seen and set aside.

7.3 Reading the Ledger

The sold record. Seven confirmed frame purchases and three confirmed frame sales are documented in the author’s private record. The first three frames arrived as single-board units, so configuration cannot explain any internal spread; the visible variable is geography, with one UAE purchase pair and five Japan purchases. On these data the purchase record is an anecdote, not a curve [confidence: low], and the author’s caution stands: the Japan purchases may be lucky grabs, and low prices at n-of-few generalize nothing.

The Japan-US asymmetry, on publishable figures. The one publishable Japan-domestic price point is a Tokyo dealer’s posted ¥13,200 for a loose BKAM-101 — the scarcest board in the family (\$7.4) — while the one live third-party US-visible frame ask is \$999.99. Two prices from two venues make no market model [confidence: low]; they do establish that, as of July 2026, the cheap observed prices were Japanese and the expensive observed prices were American, with no overlap in the documented record.

The ask. The third-party \$999.99 listing sits far above every confirmed frame transaction in the record, purchases and sales alike [comparison verified against the author’s private figures]. Nothing in the documented record supports it converting at that level; it has not sold as of this drafting and should be read as an aspiration, not a valuation. At fleet level the record supports three statements: a working AMS-100 has repeatedly changed hands for materially less than the standing ask; the sole outlier ask has no demonstrated buyer; and the Japan-domestic price points observed point against valuing units at the ask’s end of the range.

The one confirmed module transaction. The BKAM-102 lot is the only confirmed per-board transaction known for any BKAM module, and it is a **NOS** transaction — the boards arrived factory-sealed in Sony cartons (\$4.2). It demonstrates that the 102

is the board that circulates, but the circulation mechanism documented is surviving unsold stock, not decommission pulls; the pull mechanism remains inferred from the UAE strip pattern (§4.4), not evidenced by any transaction. Whether a NOS board commands a premium over a pull cannot be tested on one data point. The BKAM-101's loose market consists of one listing record; the BKAM-103's, of asks and claims only.

7.4 Scarcity by Component

The four documented SKUs occupy sharply different scarcity positions. The configuration logic — which boards get pulled, which stay aboard, and why — is treated at §3.6, and the registry's as-surfaced and serial censuses live at §4.4; this section states only what the market record adds.

AMS-100 frame — obtainable, thinly traded. Seven frames were purchasable by one collector within one year across two geographies (§4.1), the July 2026 sweep documented further dealer and auction inventory in Japan (§7.6), and two of the author's frames found buyers. This argues against genuine rarity of the frame. What is rare is *transactions in visible channels*: US-visible channels clear a handful of units per year at most [confidence: low; visibility is the limiting factor], while the Japan-domestic picture is a point-in-time snapshot supporting no rate claim in either direction.

BKAM-102 — the liquid part. The only module with a documented loose market and a confirmed transaction. Two feeds are documented or inferred: AES/EBU was the least perishable transport in the plants that ran these units (§3.3), so 102s stayed in service longest and were worth pulling for reuse [inferred]; and unsold new stock survives sealed — the one documented lot is NOS (§4.2) [documented, n = 1 lot].

BKAM-103 — present but priceless. The board most likely to remain aboard a surplus frame in the channels where audio boards are harvested (§3.6, §4.4), yet no confirmed standalone sale is known; the loose-103 record is a Korean eBay listing and an export dealer's unverified stock claim. Inert without an SDI plant (§5.4), it generates no independent demand and therefore no independent confirmed price: the 103 trades as frame ballast.

BKAM-101 — scarce as a board in every visible market. The market reading of the 101 census (§4.4, §3.2) is this:

- **Confirmed per-board transactions: zero.** No standalone BKAM-101 sale has ever been confirmed. Every working 101 in the author's custody arrived inside a unit-level purchase — a confirmed transaction *containing* a 101, not a 101 transaction — and the ledger keeps that distinction.
- **Observed listings: one** — a Tokyo dealer's loose board at a posted ¥13,200, sold out, dates unknown (§7.2). The only per-board price datum of any kind.
- **Supply side:** every physically verified working 101 known to the record came out of Japan-domestic supply (§7.6). Beyond the author's, the record holds the Tokyo dealer's board and the EXT-02 unit in Los Angeles (report tier; non-working, offered then withdrawn, owner retains — §4.5.2), the only reported failed example.

The scarcity thesis is channel-scoped, not weakened: **in US-visible channels the void is intact** — no US listing of a 101 has ever been observed, and the author’s 2026 want-to-buy thread went unfilled — while in Japan-domestic channels the board reached one collector repeatedly within a year, which is strong evidence that it exists there in some quantity and no evidence about how much [point-in-time, 2026; channels of unknown depth; no repeatability claim]. The market half of the thesis — that the 101 cannot be bought *as a board* anywhere visible — is untouched. Whether surviving 101s exist in quantity in still-installed analog monitor points, in Japanese private and dealer stock, or were largely consumed by attrition cannot be determined from the present record [confidence: low-to-moderate; the US-visible absence remains evidence of absence per §3.2, channel-scoped].

7.5 Structural Scarcity: The Sony Spares Policy

Section 6.1 documents a service-policy fact with direct market consequences: for the three option boards its manuals cover, Sony did not stock bare PCBs. The prescribed remedy for a failed board was **whole-product exchange** — AL-42 → BKAM-101, DAC-37 → BKAM-102, SDI-43 → BKAM-103 [MM, cited via §6.1; not re-derived here; the later HD-SDI board of §3.1 postdates the manuals and is not covered].

Two consequences follow. First, there was never a Sony repair-parts channel dispensing bare boards — but the boards were themselves boxed products, so the aftermarket can be seeded from two pools: decommission pulls and surviving unsold new stock. The stronger inference that every loose board in circulation is a pull is excluded by the registry’s own module lot, which arrived factory-sealed (§4.2): the NOS pool exists. Second, and most acutely for the BKAM-101: a failed 101 had no repair path through Sony parts, so attrition of the analog board was permanent from the day of failure. The 2026 market inverts Sony’s model — circulating loose boards, pulls or NOS, *are* the de facto service spares (§6.1) — but only for the board types that circulate. For the 101 the factors compound: no repair path, no documented pulls, no documented surviving new stock, and exactly one observed loose example anywhere, sold out on sighting. [Policy: MM. The compounding argument: author.]

7.6 Supply Channels

Five channels are visible in the record, in descending order of evidence.

NOS / dealer-stock liquidation (boards). The documented module lot — three factory-sealed BKAM-102s in Sony cartons, sold by a US electronics recycler (§4.2) — shows that unsold new stock of the option boards still surfaces, a channel categorically distinct from decommission surplus. Where the seller’s stock originated is unknown; the cartons’ B-AVL stock stickers are the only lead [channel existence documented, n = 1 lot; origin unknown].

Gulf decommission surplus. The UAE pair (11341, 10010) arrived stripped to a lone BKAM-103 — the signature of plants harvesting reusable audio boards before surplussing frames — and wearing tamper seals with exactly consecutive serials, the mark of a single handler batch-processing the pair (§4.4). Two units from one region is a pattern-shaped observation, not a finding [confidence: low-to-moderate], but it is

the channel hypothesis the as-surfaced configurations best fit, and the seals identify a handler's existence, not the handler [\$4.4; single-issuer moderate-to-high; issuer unknown].

Japan professional-video dealers. The July 2026 sweep identified two Tokyo-area professional-broadcast dealers holding or having held AMS-family stock — unnamed by author ruling. One has carried a 102-equipped, function-tested AMS-100 [dealer listing; stock status unrecorded]; the other's sold-out loose BKAM-101 is \$7.2's listing record. This is a channel class categorically distinct from auction surplus — condition-noted, function-tested, dealer-warranted inventory, broadcast-native. Two dealer listings constitute the entire record [channel existence documented; depth unknown; point-in-time, July 2026]. An adjacent, weaker datum: an Asian export dealer's unverified claim of five 103-equipped units in stock, which if real would extend the class beyond Japan [dealer claim — unverified].

Japan electronics-recycler surplus. The author's five Japan frames came through a single electronics-recycler channel (\$4.1). Frame 12793 arrived carrying a lone BKAM-102 — the board the pull economics predicts would be harvested — and no 103; 11353 arrived complete (101 + 103, the only 103 from this channel); 11878, 12886, and 13048 arrived 101-only, 2×101 , and 101-only [author statement]. Every frame carried at least one board with a standalone use and two arrived complete: Japan recycler supply looks different from the Gulf channel, not gentler, and it is the documented source of every working 101 in the record [confidence: low-to-moderate; $n = 5$, one channel]. The sweep added a counter-observation in the same channel: a Yahoo Auctions JP frame listed 103-only with slot B blank [auction listing, date unrecorded], which *does* match the Gulf strip signature — the pull pattern is channel-dependent, not geography-exclusive (§3.6). Japan remains the rational hunting ground for the outstanding Operation Manual, with the two dealers above as direct-inquiry candidates.

US hi-fi resale. The US hi-fi reseller's sale (≈ 2017) shows at least one unit passed through a consumer hi-fi channel rather than broadcast surplus. One event, price unknown; it establishes only that the unit has occasionally been marketed *as audio gear*.

7.7 Demand Signals

Visible demand consists of: the author (seven purchases, one public want-to-buy thread); whoever bought the author's two 102-only frames [author statement; particulars withheld]; whatever buyer cleared the US hi-fi reseller's unit in 2017; whatever buyer, if any, cleared the Tokyo dealer's loose 101 (a sold-out page implies demand met supply, but confirms neither party nor price); and the absence of anyone out-bidding or pre-empting the author's 2026 purchases. The \$999.99 lister evidently believes a premium buyer exists; that listing's persistence is the one visible test of what the US market will bear.

The unfilled want-to-buy thread is a **venue finding, not an inventory finding**. A public, specific, funded request for BKAM-101-equipped units (2026) produced nothing in over a month while, unknown to the author at the time, precisely that configuration was purchasable in Japan and a loose 101 had been inventoried by a Japanese dealer.

The demand signal never reached the channels holding supply; the thread’s silence is not evidence of global absence [confidence: low-to-moderate; single thread, single venue; the Japan supply picture is itself point-in-time].

7.8 Summary Position

As of this writing (2026), the market position is this:

The frame is obtainable, cheaply traded, and unpriced. The AMS-100 has essentially no collector market, no literature, and no price consensus. In US-visible channels the record is nearly silent and nearly all one collector’s; the sole third-party ask, \$999.99, sits far above every confirmed transaction. In Japan-domestic channels a documented July 2026 snapshot shows dealer and auction inventory at posted prices — where publishable, a Tokyo dealer’s ¥13,200 loose 101 — far below the US-visible ask, and the author’s acquisitions through a Japan electronics recycler are a second dated observation of the same geography. These are point-in-time findings and support no claim about ongoing availability.

Of the boards, only the BKAM-102 has a confirmed per-board market — one documented lot transaction, sealed NOS (\$4.2). The BKAM-103 travels as frame furniture, its loose record limited to asks and claims.

The BKAM-101 is the scarce object in the family as a board, not as an installed part. Zero confirmed per-board transactions anywhere; one listing record; a void intact in US-visible channels; unfilled public demand. Every physically verified working example known came out of Japan-domestic supply (\$4.4, \$7.4). Any confirmed per-board BKAM-101 transaction would be the single most valuable market datum this section could acquire; none is sought.

Sources for this section: the author’s private records (Section 4) for all confirmed author transactions and sales; the July 2026 market-listing sweep [dated; URLs in the project record] for the Japan-domestic findings — a listings sweep, distinct from the July 2026 literature sweep, without effect on the primary-literature claim; eBay listing observation, July 2026, for the third-party US ask; a US hi-fi reseller’s archived listing (identity in the project record) for the single known third-party US sale event; the author’s 2026 want-to-buy thread for demand signals; §6.1 [MM] for the Sony spares policy. Section 4 owns the per-unit acquisition record; this section owns the asks-vs-solds ledger and the market interpretation. Bounds for this section are stated in Appendix F.

Appendix A — Specification Tables

*Source discipline: neither service manual contains a published specifications table [IM][MM], so every figure below carries its provenance: **[IM]** Installation Manual, **[MM]** Maintenance Manual (including alignment specifications, which serve as de facto electrical specs), **[author]** hardware-verified on registry units, **[dealer — un-verified]** resting solely on dealer listings (a broadcast-equipment dealer, identity in*

the project record) which only the Operation Manual could verify — that manual is not located and the search is closed (§2.8), so every such cell stands permanently at this tier. Figures are collected here from Sections 2, 3, and 6; this appendix restates, it does not re-derive. **Coverage:** the frame and the three option boards documented by Sony’s 1999 service literature. A fourth, later HD-SDI board is attested by a single photograph (§3.1) [social-media photo, unverified]; its model number is unread (published BKAM-10■) and nothing is known of it beyond its panel legend, so no table is provided for it.

A.1 Frame — General

Item	Value	Source
Model	AMS-100 Audio Monitor Speaker	plate [author]
Origin	Made in Japan	plate [IM][author]
Dimensions (W × H × D)	482 mm overall / 424 mm body × 44 mm (1RU) × 350 mm	[IM]; height [author, measured]; depth not author-measured (measurement declined)
Weight	not stated in available primary sources; author measurement of a registry unit declined by ruling	[unknown — Operation Manual is the remaining path]
Power	AC 100-240 V ±10%, 50/60 Hz, 25 W, inrush 20 A; plate rating 0.5 A	[IM]; plate [author]
Regional variants	none — universal supply; regional cord sets only (US/Canada; DK-2401 UK; DK-2401 AE)	[IM]
Operating temperature	5-40 °C	[IM]
Rack mounting	RMM-30 rail kit required (drop hazard on other rails)	[IM]
Serial applicability	“(SY) Serial No. 10001 and Higher,” frame and the three documented boards (BKAM-101/102/103); the HD-SDI board of §3.1 is outside both manuals	[IM][MM]
Option slots	2, rear; any board in any slot; unused slots blanked	[IM]
Leakage current (service limit)	≤ 3.5 mA	[MM]

A.2 Frame — Monitoring Path

Item	Value	Source
Internal speakers	2 W + 2 W stereo	[dealer — unverified]
Speaker driver part no.	1-529-353-11 (×2)	[MM]
Headphone output	−12 dBu at 8 Ω	[dealer — unverified]
Monitor output, nominal	+4 dBm (service alignment +4.0 ± 0.1 dBm), 2 ch, balanced triplets on REMOTE D-sub	[MM]; carriage [IM]
Monitor output, max / source Z	+24 dBm / 600 Ω balanced	[dealer — unverified]
Internal program standard	−10 dBs, 2-channel analog, on 48-pin DIN slot connector (all board types)	[MM]
Level meter	2-channel LED, +3 to −20 scale, 0-reference calibration	[IM]
Meter calibration	front panel (hold slot button 3 s); reference +4 dBm (BKAM-101) or −20 dBFS (BKAM-102); 10-min warm-up	[IM]
Power-on mute	yes (designed behavior)	[MM]
Front-panel controls	POWER, VOLUME, PHONES; SLOT A/B; GP 1-4; CH-L 1-4; CH-R 1-4	[MM][author]

A.3 BKAM-101 — Analog Input Board (AL-42)

Item	Value	Source
Sony assembly part no. / stock code	1-673-874-11 / S04-378T	[author, July 2026 intake — s/n 10505]
Inputs	2 ch balanced analog, XLR-3-31, with XLR-3-32 true loop-through pairs (CN1/CN2, CN21/CN22)	[IM][MM]; panel complement [author]
Nominal level	+4 dB / 0 dB, per channel (S2 ch 1 / S22 ch 2); factory +4 dB; switches internal (board removal required)	[IM][MM]; accessibility [author]
Maximum level	+24 dBm	[dealer — unverified]

Item	Value	Source
Input impedance	10 k Ω bridging; 600 Ω termination switchable per channel (S1 ch 1 / S21 ch 2), panel-mounted toggles, legended ON/OFF	bridging [dealer — unverified]; termination function [author confirmed — functional test, July 2026] ;
Alignment	RV1/RV21 per channel → 347 $\pm$ 3 mV at CN101 pins 19/35 (–10 dBs internal standard); trimmers 4.7 k Ω	accessibility [author] [MM]; trimmer value [author]
Op-amp complement	JRC 4560DD $\times$ 2 (IC1/IC2); A5201A $\times$ 2 (IC3/IC4)	[author, n = 1 — s/n 10505; the four later Japan-recycler boards were not opened]
Protection	IC links PS101/PS102, Sony part 1-533-984-11	[MM]; presence [author]
Serial marking	five-digit “SERIAL NO.” PCB label, component side; panel bare	[author, n = 1 for the PCB label; panel-bare n = 5 at photo tier]
Functional status	every registry example working on author hardware; census and placements at \$4.2	[author — functional; 600 Ω exercised electrically on 10505 only]
Spares policy	not stocked as bare PCB; whole-product exchange (AL-42 → BKAM-101)	[MM]

A.4 BKAM-102 — AES/EBU Digital Input Board (DAC-37)

Item	Value	Source
Sony assembly part no. / stock codes	1-673-875-11 / S92-042A and S96-547A (two examples, codes differ — \$6.4)	[author, s/n 10099]; [contributor photo, EXT-01, s/n 10016]
Input	1 AES/EBU stream (2 ch): XLR-3-31 (110 Ω) or BNC (75 Ω), selected by board switch S101	connectors [IM]; S101 [MM]; 110 Ω corroborated by termination R102 [MM]; 75 Ω figure [dealer — unverified]

Item	Value	Source
Configuration	S101 XLR/BNC input select — module-panel toggle, operable in situ (no auto-detect, no frame front-panel indication); S102 position on layout, not mounted [MM — populated-or-not on s/n 10099 unresolved, do not infer]	[MM]; accessibility [author — §3.3]
Lock indication	green rear LED D102 (AES/EBU input lock)	[MM]
Decode chain	CS8412 receiver → AK4320 D/A; CPLD at IC101 (part/firmware varies by example: EPM7064S V1.10 / EPM7032S V1.20 — flagged discrepancy, §6.4)	[MM]; CPLD [author + contributor photo]
Alignment reference	1 kHz at −20.0 dBFS, 48 kHz, emphasis off → +4.0 ± 0.1 dBm at monitor output	[MM]
Protection	IC links PS101-PS103, Sony part 1-533-282-21	[MM]
Serial marking / packaging	five-digit “SERIAL NO.” PCB label, component side (present on sealed-NOS stock — §4.2 demonstrated); same serial expressed on carton for sealed stock; panel bare; carton part no. P-96604680-L	[author, s/n 10099 + author carton photograph, held unpublished; §4.2]
Spares policy	whole-product exchange (DAC-37 → BKAM-102)	[MM]

A.5 BKAM-103 — SDI De-embedder Board (SDI-43)

Item	Value	Source
Sony assembly part no. / stock codes	1-673-876-11 / S94-510A and S06-091A (two examples, codes differ — §6.4)	[author, s/n 10631]; [contributor photo, EXT-01, s/n 10130]
Input/output	serial digital video, SDI IN + SDI OUT (loop-through), 75 Ω BNC; panel legend marks the loop as a driven output (buffer symbol; drive-vs-reclock not determinable from the legend)	[IM]; panel legend [author]
De-embedding	up to 16 audio channels; 4 groups $\times$ 4 channels; GP buttons select group, CH-L/CH-R buttons select channel per monitor leg	count [dealer]; architecture [MM]
Switches	S101 free-run adjust (factory OFF); S102 video system 525/625 (factory 525); S103-1 serial format D1/D2 (factory OFF = D1); S103-2/-3/-4 factory use — all internal, board removal required	[IM]; accessibility [author]
Free-run alignment	S101 ON; TP102; RV102 $\rightarrow$ 26.9 $\pm$ 0.1 MHz (D1) / RV101 $\rightarrow$ 14.2 $\pm$ 0.1 MHz (D2); restore S101 OFF, S103-1 to D1	[MM]
Documentation erratum	alignment text “S301-1” is a typo for S103-1 — board silkscreen reads S103	[MM; author determination, §6.3; hardware-corroborated, author photo, July 2026]
Lock indication	green rear LED D104 (SDI input lock)	[MM]
Decode chain	CXB1342R serial decoder $\rightarrow$ CXD8280AQ time-base expansion (group de-embed) $\rightarrow$ AK4320 D/A; CPLDs EPM7096 (IC105) + EPM7064S (IC102), firmware states vary by example (flagged — §6.4)	[MM]; CPLDs [author + contributor photo]
Protection	IC links PS101-PS103, Sony part 1-533-282-21	[MM]

Item	Value	Source
Serial marking	five-digit “SERIAL NO.” PCB label, solder side (both examples); panel bare (both examples)	[author, s/n 10631]; [contributor photo, s/n 10130]
Spares policy	whole-product exchange (SDI-43 → BKAM-103)	[MM]

A.6 Frame Board Set

Board	Function	Notable components	Source
AU-265	main: CPU, slot switching, amplification, monitor drivers	HD64180ZR CPU; IC202/203 slot select; IC204/205 monitor drivers; IC201 speaker amp (TA8200AH/TA8216H, dual designation as recorded)	[MM]
MT-136	level meter	BA682A drivers; 2× DC-3H7GWA LED arrays	[MM]
HP-97	headphone amplifier	NJM4556AD; RV1 level pot; mute logic	[MM]
SW-10	front-panel illuminated buttons	dynamic scan drive	[MM]
VR-257	volume control	20 kΩ / 20 kΩ pot	[MM]
PSU	switching regulator	—	[MM]

Maintenance modes and error blink codes are not duplicated here; §6.2 is the reference. Mating-connector specifications (ITT Cannon XLR-3-12C/-11C; 75 Ω BNC on 5C-2V, ≤ 200 m) are at §2.5. Bounds for this appendix are stated in Appendix F.

Sources: Sony Installation Manual [IM]; Sony Maintenance Manual [MM]; author hardware verification on registry boards s/n 10505 / 10099 / 10631 and registry frames (Sections 2, 3, 6; author photographs held in the project record, not published — Appendix C); EXT-01 contributor photographs, boards s/n 10016 / 10130 (plates C-12-C-15) [contributor photo, EXT-01]; the broadcast-equipment dealer’s listings for every cell flagged [dealer — unverified].

Appendix B — REMOTE Connector: Signals, Encoding, and Pin Assignment

B.1 Connector

Rear-panel **REMOTE**: female D-sub 25-pin [IM; MM §1-6 corroborates]. Carries the frame's two balanced monitor-output channels and the full remote select/status interface (§2.5). No dedicated Sony remote panel accessory for this connector has been identified [confidence: absence of evidence only, §2.5].

B.2 Signal Complement

Signal	Direction	Function	Source
MONITOR OUT 1 (hot / cold / gnd)	out	monitor channel 1, balanced, +4 dBm nominal	[IM][MM]
MONITOR OUT 2 (hot / cold / gnd)	out	monitor channel 2, balanced, +4 dBm nominal	[IM][MM]
CH SEL L1 / L2	in	2-bit code: left-leg channel select (1-4) — code table at B.4.1	[IM][MM]
CH SEL R1 / R2	in	2-bit code: right-leg channel select (1-4) — code table at B.4.1	[IM][MM]
GP SEL 1 / 2	in	2-bit code: embedded-audio group select (1-4) — code table at B.4.1	[IM][MM]

Signal	Direction	Function	Source
REM ENABLE 1 / 2	in	remote-enable, functions differentiated [MM §1-6, Remarks]: ENABLE 1 (pin 25) — key operations of the unit are inhibited; only key operation from the remote control unit is enabled. ENABLE 2 (pin 22) — only the SLOT select command from the remote control keys is enabled.	[IM][MM]
STS outputs (SEL counterparts)	out	status lines mirroring each select state	[IM][MM]
REM-A / REM-B	in	SLOT select command [MM §1-6, Remarks]. Bounds of the record: the per-line slot mapping (REM-A → SLOT A) is name-implied only — the MM does not state it and it is not recorded as data; the encoding of the two command lines is likewise unstated.	[MM]

Signal	Direction	Function	Source
STS-A / STS-B	out	SLOT status output [MM §1-6, Remarks]; corroborated by MM §1-5-1, which routes the SLOT A/B select-button status to pins 14/15 and identifies these as the lines governed by the level/pulse type configuration (B.3). Same per-line mapping bound as REM-A/B applies.	[MM]
Ground(s)	—	signal/chassis returns — one common GND (pin 7) plus per-output grounds (pins 21/23)	[IM][MM]

B.3 Electrical Behavior

- All remote inputs and outputs are TTL-compatible, active low — MM verbatim: “The inputs and outputs are TTL compatible (L-ACTIVE)” [MM §1-6].
- Paired SEL lines carry a **2-bit H/L code** selecting among the four groups or four channels [MM]. The code-to-selection table is transcribed verbatim at B.4.1 [MM §1-6, footnote (*1)]; it does not appear in the IM.
- **STS output type is frame-resident configuration:** MM §1-5-1 states the level/pulse selection for **the SLOT status signal on pins 14/15 (STS-A/STS-B) only** — either continuous level or a **500 ms active-low pulse**, set via the maintenance modes GP4+CH-R3 / GP4+CH-R4 at power-on (§6.2) [MM §1-5-1]. Whether the other STS lines share the configured type is **unstated in the MM** — open, Operation Manual or bench probe. The registry frames’ configured state is unrecorded (Appendix F).
- Group/channel selections made via REMOTE mirror the front-panel GP and CH-L/CH-R buttons (§3.4) [MM].

B.4 Pin Assignment [IM — transcribed verbatim]

Transcribed pin-for-pin from the Installation Manual’s §1-5 pin-assignment table (D-sub 25-pin, female; the IM’s external-view drawing orients the connector with

pins 13→1 on the upper row and 25→14 on the lower) [IM]. Signal names are reproduced as printed, including the IM’s own punctuation inconsistencies (e.g., “REM-CH.SELECT” vs “STS-CH. SELECT”); no name has been normalized.

Pin	Signal	Pin	Signal
1	REM-A IN	14	STS-A OUT
2	REM-B IN	15	STS-B OUT
3	REM-CH.SELECT L1 IN	16	STS-CH. SELECT L1 OUT
4	REM-CH.SELECT L2 IN	17	STS-CH. SELECT L2 OUT
5	REM-CH.SELECT R1 IN	18	STS-CH. SELECT R1 OUT
6	REM-CH.SELECT R2 IN	19	STS-CH. SELECT R2 OUT
7	GND	20	STS-GROUP SEL. 1 OUT
8	MONITOR OUT 2(H)	21	MONITOR OUT 2(G)
9	MONITOR OUT 2(C)	22	REM-ENABLE 2 IN
10	MONITOR OUT 1(H)	23	MONITOR OUT 1(G)
11	MONITOR OUT 1(C)	24	STS-GROUP SEL. 2 OUT
12	REM-GROUP SEL. 1 IN	25	REM-ENABLE 1 IN
13	REM-GROUP SEL. 2 IN		

Reading notes, at transcription tier only: the layout is not mirror-symmetric — remote inputs and their STS counterparts sit at offset positions (e.g., GROUP SEL inputs at 12/13 but their status outputs at 20/24), and the two REM-ENABLE inputs are split across the rows (25 and 22). Anyone building a cable should work from this table alone, not from an assumed symmetry. The monitor outputs appear as full balanced triples: hot/cold on 8/9 and 10/11, with dedicated grounds on 21 and 23 in addition to the common ground on 7 [IM].

B.4.1 Select-Code Truth Table [MM — transcribed verbatim]

Transcribed from the Maintenance Manual §1-6, footnote “(*1)” to the REMOTE pin-assignment table [MM]:

	CH/GP1	CH/GP2	CH/GP3	CH/GP4
SEL-1	H	L	H	L
SEL-2	H	H	L	L

Scope, per the MM’s own (*1) markings: one code, all six selector pairs — the command inputs CH-L SEL-1/2 (pins 3/4), CH-R SEL-1/2 (pins 5/6), and GP SEL-1/2 (pins 12/13), and the status outputs CH-L SEL-1/2 (pins 16/17), CH-R SEL-1/2 (pins 18/19), and GP SEL-1/2 (pins 20/24) [MM §1-6]. No interpretation is layered on the table — in particular, no reading of the H/H state for selection 1 against the interface’s stated L-ACTIVE convention is adopted. The code is reproduced as printed and should be wired as printed.

B.4.2 Cross-Manual Corroboration

The MM carries its own REMOTE pin table (§1-6), which matches the IM table above **pin-for-pin, 25 of 25**, and adds two columns the IM lacks: an I/O direction column (consistent with the IN/OUT suffixes throughout) and a Remarks column (the source of the REM-A/B, STS-A/B, and REM-ENABLE function resolutions in B.2) [MM §1-6]. Signal naming differs typographically only — the IM prints “REM-CH.SELECT L1”, the MM “REM-CH-L SEL-1”, and so on; no functional divergence at any pin. B.4’s IM transcription stands as printed, unnormalized; the MM table is an independent corroboration of it, not a replacement. One MM-internal naming variance is noted for bench users: the alignment section’s connection footnote (§2-1-3) calls pins 10/11/23 and 8/9/21 “AUDIO OUT 1/2” where §1-6 calls the same pins “MONITOR OUT 1/2” — identical pin assignments, different label [MM].

B.5 Mating Hardware

REMOTE mate: **D-sub 25-pin, male — Sony part 1-566-356-11** [IM, §1-6]. ITT Cannon XLR-3-12C (male) / XLR-3-11C (female) or equivalent for board XLR positions; 75 Ω BNC on 5C-2V coaxial (Fujikura or equivalent), maximum 200 m run, for board BNC positions [IM].

Standing bounds of this appendix (no transcription items remain): the per-line REM-A → SLOT A / STS-A → SLOT A mapping is name-implied, not data — movable only by the Operation Manual or a bench probe; whether the level/pulse type extends beyond STS-A/STS-B is unstated [MM §1-5-1 scope]. Open items are consolidated in Appendix F.

*Sources: Sony Installation Manual [IM], §1-5 pin-assignment table and §1-6 mating-connector data (part no. 3-202-309-01, 1st Ed.); Sony Maintenance Manual [MM], §1-5-1 (STS output type), §1-6 (REMOTE pin table, Remarks column, footnote (1)), §2-1-3 (alignment connection footnote) (part no. 9-967-830-01, 1st Ed.).**

Appendix C — Photo Plates (caption record)

*Plates C-10-C-21: contributor photographs, EXT-01 intake of July 2026 — cited [**contributor photo, EXT-01**]; twelve plates selected from the 22-image packs (designations EXT01-P1-01...09 / P2-01...13; the export path stripped the pack filenames, so the designations are content-verified rather than filename-verified). The units photographed by EXT-01 are **not registry members** (§4.5 scope wall): the AMS-100 shown is frame 10047, and plates C-19-C-21 document the two Japan-domestic AMS-3 units of Appendix E. Publication of the EXT-01 images is by author ruling; contributor credit in Appendix D. Plates C-22-C-23: archived seller-listing photographs, the as-received documentation of record for frame 11353. Plates C-35-C-40: author photographs of 2026, logged and captioned by author ruling — the final configurations of frames 13048 and 12886, and the four-plate documentation of the AMS-3 export pair of Appendix E. (Drafting note: C-35 and C-40 show registry hardware and are logged as caption entries rather than withdrawn, on the ruling that logged them; the*

earlier withdrawal ruling enumerated a fixed set of plates and is not extended here by inference. Their images publish no more than any other plate's — none do.)* **Plates C-1-C-9 and C-24-C-34 are withdrawn by author ruling (August 2026)**: the author's own photographs of registry hardware — frame rears, plates, seals, the lot cartons, pulled boards, and the fleet — are not published. The photographs remain in the project record, and every reading that rested on them stands at **[author]** tier on that record; where a section cites a withdrawn plate number, the reference resolves here to a withdrawn entry, not to an image. Frames are identified by frame serial throughout (the drafting record numbers map to serials in Appendix D). **No image is reproduced in this monograph** (author ruling, August 2026): every plate publishes as a caption — the content-verified description of record — and the plate log records which images exist, at what tier, and where they are held. Configurations described in any plate are as of its date; current configurations are in §4.1.*

Plate log

The log is the mapping of record between plate number, source, and holding. Plate numbers are stable: withdrawn and image-less entries are retained, not renumbered, so that plate references across Sections 1, 4 and 6 and Appendices A, D, and E remain valid.

Plate	Source	Subject	Date	Tier	Holding
C-1	author photograph (project record)	11341 (upper) + 12793 (lower), rears	—	[author]	withdrawn (author ruling, August 2026)
C-2	author photograph (project record)	plates 11341 / 12793	—	[author]	withdrawn (author ruling, August 2026)
C-3	author photograph (project record)	BKAM-103 panels 0128521 / 0105241	—	[author]	withdrawn (author ruling, August 2026)
C-4	author photograph (project record)	seal C094183, 11341	—	[author]	withdrawn (author ruling, August 2026)
C-5	author photograph (project record)	BKAM-102 lot cartons	—	[author]	withdrawn (author ruling, August 2026)

Plate	Source	Subject	Date	Tier	Holding
C-6	author photograph (project record)	10010 rear, 2× 102	—	[author]	withdrawn (author ruling, August 2026)
C-7	author photograph (project record)	plate 10010, label 0105235	—	[author]	withdrawn (author ruling, August 2026)
C-8	author photograph (project record)	seal C094182, 10010	—	[author]	withdrawn (author ruling, August 2026)
C-9	author photograph (project record)	10010 front	—	[author]	withdrawn (author ruling, August 2026)
C-10	EXT01-P1-04 → C-10	10047 rear, two-board, over AMS-3	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-11	EXT01-P1-09	plate 10047	July 2026	[contributor photo, EXT-01]	original with contributor, not supplied; caption only
C-12	EXT01 pack image → C-12	BKAM-103 open board, panel-forward	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-13	EXT01-P2-07	BKAM-103 CPLDs / oscillator	July 2026	[contributor photo, EXT-01]	original with contributor, not supplied; caption only
C-14	EXT01-P2-06 → C-14	BKAM-103 solder side, SERIAL NO. 10130	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only

Plate	Source	Subject	Date	Tier	Holding
C-15	EXT01-P2-09 → C-15	BKAM-102 component side, SERIAL NO. 10016	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-16	EXT01-P2-12 → C-16	BKAM-102 panel, BNC→RCA adapter	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-17	EXT01-P2-13 → C-17	BKAM-102 solder side, B-STOCK	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-18	EXT01-P2-10	10047 slot bay, 48-pin DIN	July 2026	[contributor photo, EXT-01]	original with contributor, not supplied; caption only
C-19	EXT01-P2-01 → C-19	AMS-3 pair, fronts	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-20	EXT01-P2-02 → C-20	AMS-3 pair, rears	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-21	EXT01-P1-05 → C-21	AMS-3 10685 plate	July 2026	[contributor photo, EXT-01]	contributor image, project file; caption only
C-22	0219174003_69965363821fb7026	as listed, 101 = A + 103 = B	July 2026 (capture)	[seller listing photo, archived]	author archive; caption only
C-23	0219174103_6996539866092026	detail as listed	July 2026 (capture)	[seller listing photo, archived]	author archive; caption only (custody note at plate)
C-24	author photograph (project record)	plate 11353	—	[author]	withdrawn (author ruling, August 2026)

Plate	Source	Subject	Date	Tier	Holding
C-25	author photograph (project record)	BKAM-101 10505 panel	—	[author]	withdrawn (author ruling, August 2026)
C-26	author photograph (project record)	BKAM-101 10505 component side	—	[author]	withdrawn (author ruling, August 2026)
C-27	author photograph (project record)	BKAM-102 10099 panel	—	[author]	withdrawn (author ruling, August 2026)
C-28	author photograph (project record)	BKAM-102 10099 component side	—	[author]	withdrawn (author ruling, August 2026)
C-29	author photograph (project record)	BKAM-103 10631 panel	—	[author]	withdrawn (author ruling, August 2026)
C-30	author photograph (project record)	BKAM-103 10631 component side	—	[author]	withdrawn (author ruling, August 2026)
C-31	author photograph (project record)	BKAM-103 10631 solder side	—	[author]	withdrawn (author ruling, August 2026)
C-32	author photograph (project record)	fleet photograph, seven frames (subject per image unconfirmed)	—	[author]	withdrawn (author ruling, August 2026)

Plate	Source	Subject	Date	Tier	Holding
C-33	author photograph (project record)	fleet photograph, seven frames (subject per image unconfirmed)	—	[author]	withdrawn (author ruling, August 2026)
C-34	author photograph (project record)	fleet photograph, seven frames (subject per image unconfirmed)	—	[author]	withdrawn (author ruling, August 2026)
C-35	author photograph (project record)	frame 13048 rear as sold — BKAM-103 (0128521) slot A + BKAM-102 slot B — stacked above AMS-3 20087 rear	2026	[author]	project record; caption only
C-36	author photograph (project record)	AMS-3 20087 model plate + institutional asset tag	2026	[author]	project record; caption only
C-37	author photograph (project record)	AMS-3 front panel, export variant	2026	[author]	project record; caption only
C-38	author photograph (project record)	AMS-3 rear panel, input and speaker-select field	2026	[author]	project record; caption only

Plate	Source	Subject	Date	Tier	Holding
C-39	author photograph (project record)	AMS-3 20144 model plate + institutional asset tag	2026	[author]	project record; caption only
C-40	author photograph (project record)	frame 12886 rear — BKAM-101 slot A + BKAM-102 slot B	2026	[author]	project record; caption only

Holding state: fourteen plates publish as captions — nine from EXT-01 images in the project file (C-10, C-12, C-14-C-17, C-19-C-21), three from EXT-01 images never supplied (C-11/C-13/C-18; originals remain with the contributor, recovery closed), two from the author’s archived seller-listing files (C-22/C-23); twenty author plates withdrawn. The C-10-C-21 mapping was verified at content level after the EXT-01 exports arrived with pack filenames stripped. No image embeds at compile.

Plate C-10 (*EXT01-P1-04* → *C-10*) — [contributor photo, EXT-01] EXT-01 AMS-100 (frame 10047) rear from above, AMS-3 below: **both slots populated — BKAM-103 in A, BKAM-102 in B** — the first two-board AMS-100 photographed anywhere (§4.5.1). Slot arrangement matches the registry convention.

Plate C-11 (*EXT01-P1-09* — *image not supplied*) — [contributor photo, EXT-01] Model plate close-up: serial **10047** (second-lowest frame known, 46 above the 10001 floor), full plate text identical to the registry plates, **no date code, no market suffix** — the fourth consecutive date-free plate (§4.4), REMOTE connector and BKAM-102 legend in frame. (*Availability: not among the exported files recovered in July 2026; the pack originals remain with the contributor, recovery closed by ruling. The caption’s reads stand at intake-record tier [contributor photo, EXT-01 — intake record]; the 10047 serial is independently visible in plate C-10.*)

Plate C-12 (*EXT01 pack image* → *C-12*) — [contributor photo, EXT-01] BKAM-103 out of frame, panel-forward with the component side visible behind the panel: SDI loop-through BNC pair, latch blocks, lock-LED aperture, populated board with backplane header in view — the first open-board photograph of any BKAM-103 (§4.5.1). (*The board designation **SDI-43**, Sony assembly part **1-673-876-11**, and PCB stock code **S94-510A** are intake-record readings from the pack [contributor photo, EXT-01 — intake record]; they are not legible in the image itself.*)

Plate C-13 (*EXT01-P2-07* — *image not supplied*) — [contributor photo, EXT-01] BKAM-103 component detail: Altera EPM7096 and EPM7064S CPLDs with Sony firmware labels **V1.00** ©**SONY99**, and the **V-24M576** crystal oscillator (24.576

MHz) bearing a 1999-family code — part of the project’s first internal date evidence (§4.5.1, §1.5; board production ran at least into 1999). *(Availability: not among the exported files recovered in July 2026; pack originals remain with the contributor — recovery closed by ruling. Reads stand at intake-record tier; the §1.5 dating evidence is unaffected — it was made at intake, not at embedding.)*

Plate C-14 (*EXT01-P2-06 → C-14*) — [contributor photo, EXT-01] BKAM-103 solder side: the five-digit Sony “**SERIAL NO. 10130**” label — the headline plate for the two-serial-system finding (§4.4): five-digit PCB labels read as the true board serials [mod-high], demoting the 7-digit barcode class.

Plate C-15 (*EXT01-P2-09 → C-15*) — [contributor photo, EXT-01] BKAM-102 component side: board designation **DAC-37** (corroborating [MM]), Sony assembly part **1-673-875-11**, stock code S92-042A, the **CS8412-CS** AES/EBU receiver physically sighted (date code 9903), and the five-digit “**SERIAL NO. 10016**” label (§4.5.1).

Plate C-16 (*EXT01-P2-12 → C-16*) — [contributor photo, EXT-01] BKAM-102 panel close-up: DIGITAL AUDIO IN XLR-3F (PUSH latch), IN lock LED, and the **captive BNC→RCA adapter left in place** on the BNC — physical evidence of consumer coax on RCA cabling into this input; second independent instance of the §5.2 field practice (§4.5.1).

Plate C-17 (*EXT01-P2-13 → C-17*) — [contributor photo, EXT-01] BKAM-102 solder side: “**B-STOCK**” sticker — dealer/refurbisher grading mark; the EXT-01 unit’s one provenance artifact (a resale channel somewhere in the board’s history; interpretation moderate — §4.5.1).

Plate C-18 (*EXT01-P2-10 — image not supplied*) — [contributor photo, EXT-01] EXT-01 slot bay with the 102 out: the two 48-pin DIN backplane connectors (consistent with the [MM] architecture). Illustrative of the **photo-tier seal negative** — no seal or residue on either slot seam; the negative itself rests on the full 22-image set, not this plate alone (§4.5.1). *(Availability: not among the exported files recovered in July 2026; pack originals remain with the contributor — recovery closed by ruling. The seal negative is unaffected: it was recorded against the full pack at intake.)*

Plate C-19 (*EXT01-P2-01 → C-19*) — [contributor photo, EXT-01] The AMS-3 pair, fronts stacked: dual illuminated analog VU meters (−20...+3), SP LEVEL rotary pots, internal speaker grilles; rack ears fitted on 10685, absent on 11558 (Appendix E).

Plate C-20 (*EXT01-P2-02 → C-20*) — [contributor photo, EXT-01] The AMS-3 pair, rears: plates **11558** and **10685**, per-channel XLR loop-through input fields with IMP (600Ω/10kΩ/HIGH) and LEVEL (8/4/0 dB) slides, EXT SPEAKER 8Ω pairs, INT/EXT routing (Appendix E).

Plate C-21 (*EXT01-P1-05 → C-21*) — [contributor photo, EXT-01] AMS-3 rear close-up, unit 10685: model plate — **AC 100 V 50/60 Hz 150 W** (Japan-domestic rating vs. the AMS-100’s universal supply), plate part 2-356-601-01, **no date code** — with bilingual caution label 3-703-043-21 and input-switch detail (Appendix E).

Plate C-22 (*0219174003_6996cc63821fb.jpg — author archive*) — [seller listing photo, archived] Frame 11353 full rear, oblique from above, as listed by the seller: **BKAM-101 in slot A, BKAM-103 in slot B** — the A/B legend legible at the module

seam — REMOTE, model plate, AC inlet. The as-received arrangement's documentation of record (§4.3): the boards were removed before author photography, so this archive frame is what the slot-assignment reading rests on. (*Frame since sold.*)

Plate C-23 (*0219174103_6996cc9fb6609.jpg — author archive*) — [seller listing photo, archived] Frame 11353 rear detail, closer oblique: BKAM-101 panel (per-channel 600 Ω ON/OFF toggles, CH-1/CH-2 XLR loop-through pairs) and BKAM-103 panel, A/B legend. (*Custody note: this filename's hash differs from the project's capture record, which lists a "front" as the second file; the discrepancy is recorded and stands unreconciled — Appendix F.*)

Plate C-35 (*author photograph*) — [author photo] Two units stacked, rears. **Upper: AMS-100 frame 13048** in the configuration in which it was sold — **BKAM-103 bearing barcode label 0128521 in slot A** (SDI IN loop-through BNC pair) and **BKAM-102 in slot B** (DIGITAL AUDIO IN, BNC + XLR, IN indicator), the A/B legend legible at the module seam; model plate NO. 13048, REMOTE D-sub, ground post, AC inlet. The record of the frame's final configuration (§4.1, §4.3) and a close-range read of label **0128521**, which resolves the divergent reading carried from the fleet intake in favour of the file. **Lower: AMS-3 unit 20087**, rear, export variant (Appendix E). (*Both units since sold.*)

Plate C-36 (*author photograph*) — [author photo] AMS-3 unit 20087, rear detail: model plate — **AC 100/120/220/240 V 50/60 Hz 150 W**, plate part **2-356-602-02**, NO. 20087, "Sony Corporation / Made in Japan," **no date code** and no Japanese corporate line — with the institutional barcode asset tag **TVBE-51326** alongside, and the AC inlet (§E.2, §E.2.1).

Plate C-37 (*author photograph*) — [author photo] AMS-3 front panel, export variant, light grey finish: POWER rocker with separate indicator window, two illuminated analog VU meters (INPUT LEVEL L/R, -20 to +3 with percentage-modulation scale, red above 0), two rotary SP LEVEL pots (0-10), twin speaker grilles, legend AUDIO MONITOR SPEAKER AMS-3 (§E.3).

Plate C-38 (*author photograph*) — [author photo] AMS-3 rear panel, export variant, left and centre field: SPEAKER SELECT INT./EXT. slide; EXT. SPEAKER R/L XLR pair (8 Ω); per-channel INPUT R and INPUT L fields, each with IMP (600 Ω / 10 k Ω / HIGH) and LEVEL (8 / 4 / 0 dB) slides above an **XLR-32F + XLR-31F loop-through pair** (ITT Cannon); ground post; bilingual caution label 3-703-043-21; at right, the covered rotary component whose identification is left open at §E.3.

Plate C-39 (*author photograph*) — [author photo] AMS-3 unit 20144, rear detail: model plate identical in every field to C-36 except the serial — NO. 20144, plate part 2-356-602-02, four-voltage rating, no date code — with institutional barcode asset tag **TVBE-51320** (§E.2, §E.2.1).

Plate C-40 (*author photograph*) — [author photo] AMS-100 **frame 12886**, rear, oblique: **BKAM-101 in slot A** (CH-1 IN / CH-2 IN XLR loop-through pairs, 600 Ω ON/OFF toggle) and **BKAM-102 in slot B** (DIGITAL AUDIO IN, BNC + XLR, IN indicator), A/B legend at the seam; model plate NO. 12886, REMOTE D-sub. The record

of this frame’s reconfiguration away from its as-surfaced 2× 101 arrangement (§4.3), and the registry’s only documented 101 + 102 layout.

Sources: seller listing photographs archived by the author (C-22-C-23); EXT-01 contributor photo packs, July 2026 (C-10-C-21) [contributor photo, EXT-01], credit at Appendix D; the author’s photographs of 2026 (C-35-C-40) [author photo]; Sony Maintenance Manual [MM] for the board designations corroborated in captions. Bounds for this appendix are stated in Appendix F.

Appendix D — Sources and Evidence Record

D.1 Primary Sony Documentation

Document	Identity	Status	Obtained
Installation Manual [IM]	AMS-100/BKAM-101/102/103, 1st Edition, part no. 3-202-309-01, ©1999 Sony Corporation B&P Systems Company, printed Japan 1999-02; trilingual safety matter; J and E versions	in hand	July 2026, pdf.textfiles.com STARIN manuals mirror
Maintenance Manual [MM]	AMS-100, 1st Edition, part no. 9-967-830-01, ©1999, printed Japan 1999-04, 70 pp.; covers frame and all three boards documented in 1999 (§6.1); contains no published specifications table	in hand	July 2026, elektrotanya.com

Document	Identity	Status	Obtained
Operation Manual	supplied with unit; overview and system configuration examples; authoritative target for remaining dealer-only specifications and operator procedures	not located — confirmed absent from online manual aggregators at the July 2026 sweep; search closed by author ruling (§2.8)	—

D.2 Hardware Evidence

Author (registry):

- **Photo intake** — four frames: frames 11341 and 12793, model plates, board panels, seal C094183. Author photographs, held in the project record; withdrawn from publication by author ruling (Appendix C).
- **Photo intake** — four frames: frame 10010, plate with label 0105235, seal C094182, front panel. Author photographs, held; withdrawn from publication.
- **Carton photograph**: the BKAM-102 lot as received, factory-sealed — the complete record of the cartons' markings (author confirmation). Author photograph, held; withdrawn from publication — the carton reads in §4.2 therefore rest on the held photograph and the author's retrospective confirmation.
- **Bench checks** — S101 position check on the then-installed BKAM-102s (§5.6); 1RU height measured 44 mm; converter identification (Biaze HW-US-Z1) and chain documentation (§5.2).
- **Author confirmations (verbal, no photo record; recorded as [author confirmed])** — frame 10010 chassis sweep negative; 10010's BKAM-102s bare; square-on seal reshoot declined by ruling; cartons discarded after date inspection returning negative (retrospective confirmation against discarded objects; the held carton photograph corroborates the visible faces).
- **Physical intake of frame 11353** (nine author photographs, held; withdrawn from publication; intake record in the project file): frame serial confirmed; configuration 101 (10505) + 103 (10631) confirmed against the seller's listing; board identity and five-digit serials for all three board types; board-transfer event (§4.2); lot board 10099 = carton S01-0010099 (§4.2 demonstration).
- **Late intake, 2026** (six author photographs; caption record at plates C-35-C-40): frame 13048's rear in its final two-board configuration, immediately before sale; frame 12886's rear as reconfigured; and the four-plate documentation of the AMS-3 export pair (Appendix E), including both institutional asset tags. **Author statements, same intake [author statement]**: the reconfigurations of 12793, 12886, and 13048 (§4.1, §4.3); the sale of 13048 with its BKAM-102; the AMS-3 pair's acquisition, weight, and the sale of unit 20087.

- **Fleet photo intake** (three author photographs, held; withdrawn from publication; fleet intake record in the project file): all seven registry frames photographed — rears, fronts, plates; frames 11878 / 12886 / 13048 read for the first time; pre-move configurations of 11353, 11341, 10010 recorded. **Author statements, same intake [author statement]**: current configurations and board placements (§4.1, §4.2); functional state of every BKAM-101 (§4.2); seal C094183 status (§4.4); lot board 10154 read at its move; weight/depth measurement, and close-up plates declined.

External contribution — EXT-01 (second-party; see the §4.5 scope wall):

- **Photo intake, July 2026** — two packs, 22 images (designations EXT01-P1-01...09 / EXT01-P2-01...13; intake record in the project file): one AMS-100, frame 10047, two-board (BKAM-103 slot A + BKAM-102 slot B), with open-board photographs of both option boards — the first of any BKAM module on record — and two AMS-3 units (serials 10685 / 11558; Appendix E). Twelve images selected as plates C-10-C-21. Claims from this material carry the tag **[contributor photo, EXT-01]**; the written-answer tier ([contributor report, EXT-01]) is permanently empty — the intake is closed, and provenance, price, and condition are recorded as explicit “not disclosed” states. No ledger row derives from this source, and the contributor’s location (Australia) is a custody-geography datum only, not a supply-channel datum.
- **Credit: David Law (“myretrogamehub”), Australian PVM importer** — named with the contributor’s permission, relayed July 2026. This line is the monograph’s single credit of record; in-text evidence tags remain anonymous by standing ruling. Contributed images are publishable (author ruling).

External evidence — EXT-03 (third-party social media; not a contributor):

- **Social-media photograph, August 2026** — one public post by a collector in Taiwan, anonymous, **no contact and no follow-up by author ruling**: an AMS-100 rear panel carrying an SD-SDI module (panel label BKAM-10■, final character obscured — no digit assigned) and a module legended HD SDI IN / INPUT MONITOR / IN; poster’s caption describes an HDMI→HD-SDI→BVM→AMS-100 chain. Tag **[social-media photo, unverified]** (caption: [social-media caption, unverified]), **permanent** — below contributor-photo tier (not supplied to the project; no consent). Author-held screenshots only; **not publishable, not plates** (Appendix C unchanged). Published at §3.1; no ledger row; no intake template; record closed at open.

D.3 Transaction and Provenance Records

Identifier mapping. The monograph identifies frames by frame serial. The record numbers used during drafting (R-1 ... R-7, in order of entry to the registry) map to frame serials as follows; this table is the only place they appear. The author’s private record uses its own identifiers, which are not published.

Frame serial	Record no.	Channel
11341	R-1	UAE electronics recycler (package with 10010)

Frame serial	Record no.	Channel
10010	R-2	UAE electronics recycler (package with 11341)
12793	R-3	Japan electronics recycler
11353	R-4	Japan electronics recycler
11878	R-5	Japan electronics recycler
12886	R-6	Japan electronics recycler
13048	R-7	Japan electronics recycler

- **Private records, first three registry frames (R-1-R-3)** — acquisition documentation in the author’s private record (Appendix F).
- **Module lot receipt** (US electronics recycler) — in the author’s private record.
- **Frame 11353 transaction record** — order confirmation from the Japan electronics recycler; in the author’s private record. Seller’s listing page and product photographs archived by the author (five distinct files held against six listing images; one filename’s custody discrepancy is recorded and stands — Appendix F).
- **Frames 11878 / 12886 / 13048 acquisition** — three frames from the Japan electronics recycler, 2026; in the author’s private record.
- **Sale record** — frames 11353 and 11878, each as 102 + blank; frame 13048 as 103 (0128521) + 102; all in the author’s private record.
- **AMS-3 pair (not registry members; Appendix E)** — acquisition of units 20087 and 20144, and the sale of 20087, 2026; in the author’s private record. No AMS-3 transaction enters the Section 7 ledger, which is an AMS-100 record.

D.4 Market Observations

- eBay active listing, \$999.99 (ask, not sold), third party — observed July 2026, monitored.
- **Japan market-listing sweep, July 2026** [dated; URLs in the project record]: a Tokyo used-broadcast dealer — loose BKAM-101 “Analog Audio Module,” ¥13,200 incl. tax, page marked sold out, dates unknown → \$7.2 listing record; a second Tokyo professional-video dealer’s AMS-100 (102-equipped, function-tested) [dealer listing; stock status unrecorded]; Yahoo Auctions JP frame, 103-only, slot B blank [auction listing, date unrecorded]; Asian export dealer’s claim of five 103-equipped units [dealer claim — unverified]; Korean eBay seller, one frame + two loose 103s [listing]. Literature-vacuum claim unaffected (listings ≠ literature).
- A US hi-fi reseller’s archived sold-out listing (identity in the project record) (~Nov 2017 by photo timestamps) — existence datum, price unknown.
- Stale eBay figures (\$125-127, undated, ask/sold status unknown) — seen and excluded (\$7.2).
- A collector-forum want-to-buy thread posted by the author, 2026 (2× AMS-100 with BKAM-101) — unfilled as of drafting.

D.5 Secondary Sources

- A broadcast-equipment dealer’s product listings (identity in the project record) — source of all figures flagged [dealer — unverified]; retained as the permanent basis for those cells (§2.8).
- Sony marketing copy (AMS-3 lineage claim) — encountered during research; unverified; flagged low-confidence at §1.4. The AMS-3 is physically documented in Appendix E at two tiers [contributor photo, EXT-01; author photo]; the lineage claim itself remains unverified, and four documented units have not moved it (confidence unchanged — gated on period paperwork, of which none of the four carried any).

D.6 Negative Findings of Record

- **No editorial or community literature on the AMS-100 exists** — sweep of July 2026. This monograph is, as far as can be determined, the primary literature.
- **Operation Manual absent from all online aggregators** — confirmed July 2026.
- **No date codes on any external surface, family-wide** — registry and external AMS-100 plates, the three cartons, and all four AMS-3 plates across both market variants uniformly date-free; census at §4.4, dating evidence at §1.5.
- **No RMM-30 rails, paperwork, or Operation Manual with any registry frame; every single-board frame arrived blanked** (§4.1, §4.4).
- **No tamper seal on any Japan-sourced frame or on the external frame 10047** — the C-series seals remain a UAE-package artifact; seal record at §4.4.

Sources: the documents, photographs, records, and listings enumerated above; Sony Installation Manual [IM] and Maintenance Manual [MM] as identified in D.1; the author’s private records for every transaction; EXT-01 contributor photo packs [contributor photo, EXT-01]; EXT-03 public social-media post [social-media photo, unverified]. Bounds for this appendix are stated in Appendix F.

Appendix E — The AMS-3: Predecessor Candidate

*Source discipline: this appendix documents **four** AMS-3 units at two evidence tiers. Two Japan-domestic units, serials 10685 and 11558, read from the EXT-01 photo packs of July 2026 — cited **[contributor photo, EXT-01]** — in the same Australian custody as the §4.5.1 AMS-100. Two export units, serials 20087 and 20144, acquired by the author in 2026 and read from his own photographs and handling — cited **[author]** or **[author photo]** (plates C-35–C-39). No Sony documentation for the AMS-3 is in the project’s possession, and no period paperwork accompanied any of the four units. The author’s units are **not registry members**: the registry is an AMS-100 record and stands at $n = 7$ (§4.1); no AMS-3 transaction enters the Section 7 ledger, and the AMS-3 pair generates no ledger row of any kind. The lineage claim this appendix examines therefore ends the appendix as unverified as it began — what*

changes, and it changed twice, is what the AMS-3 is known to have been as a product.

E.1 The claim, and what this appendix can and cannot do

Sony marketing copy links the AMS-100 to an earlier **AMS-3** as its predecessor. The monograph carries that lineage at **low confidence** (§1.4) because the claim has never been sourced to period Sony engineering or product literature — only to promotional text. Physical units cannot fix that: four AMS-3s on two continents confirm that the product existed, what it was, and how it compares to the AMS-100; they cannot confirm that Sony *positioned* one as the other’s successor. The instrument that would settle the lineage is a period cross-reference — an AMS-3 document naming the AMS-100, or the reverse — and none of the four units carried one. §1.4’s confidence flag is unchanged by this appendix, deliberately, and the author’s acquisition of two units does not move it: **hardware is not provenance**.

What the appendix does instead: documents the AMS-3 as hardware, and lays the two designs side by side so that the *comparative* case — continuity of design vocabulary across a generational redesign — is on the record with its evidence, tagged as observation rather than lineage proof.

E.2 The documented units — two market variants

The four units divide cleanly into two groups, and the division is the appendix’s principal finding. It was not visible when only the contributor pair was documented.

	Unit a	Unit b	Unit c	Unit d
Serial	10685	11558	20087	20144
Model-plate	2-356-601-01	identical	2-356-602-02	identical
part				
Power rating	AC 100 V, 50/60 Hz, 150 W	identical	AC 100/120/220/240 V, 50/60 Hz, 150 W	identical
Corporate text	SONY / ソニー 株式会社 / Made in Japan	identical	SONY / Sony Corporation / Made in Japan — no Japanese corporate line	identical
Date code	none	none	none	none
Caution label	3-703-043-21 (EN/FR bilingual)	identical	3-703-043-21 (EN/FR bilingual)	identical
Rack ears	fitted	absent	fitted	fitted

	Unit a	Unit b	Unit c	Unit d
Asset marking	none observed	none observed	institutional barcode tag (E.2.1)	institutional barcode tag (E.2.1)
Weight	not measured	not measured	35 lb (≈ 15.9 kg)	35 lb (≈ 15.9 kg)
Source	[contributor photo, EXT-01]	[contributor photo, EXT-01]	[author photo; weight author- measured]	[author photo; weight author- measured]

The variant finding. Four attributes move together across the two groups: model-plate part number, supply rating, corporate text, and serial band. That is four independent axes agreeing, which is a good deal more than a single differing figure would be, and the reading of record is that **the AMS-3 shipped in at least two market variants — a Japan-domestic 100 V version and a multi-voltage export version** [confidence: moderate-to-high for the two-variant conclusion; the sample is two units per side, and both export units come from a single facility, so nothing here bounds how many variants existed or in what proportion].

Serial bands. The domestic pair reads 10685 / 11558 and the export pair 20087 / 20144 — two distinct 1xxxx and 2xxxx bands, correlating with the variant split at $n = 2$ each. All four sit above 10001, the floor the Installation Manual declares for the AMS-100 family [IM]. Whether the 2xxxx band is a separate export block, a later production period, or coincidence is **not determinable from four units**, and no volume, dating, or rate inference is drawn from any of these numbers [observation only; no confidence assigned].

The **date-free plates extend the family marking-practice finding** (§4.4) backward a product generation and now across both variants: whatever Sony’s external-marking policy was for this monitoring line, every AMS-3 plate in the record follows it. The production era of these units is unbracketed by their own surfaces.

E.2.1 Institutional asset marking on the export pair

Both export units carry a white adhesive barcode label on the rear panel, beside the model plate, reading **PROPERTY OF THE CHRISTIAN SCIENCE PUBLISHING SOCIETY** with a barcode and the human-readable strings **TVBE-51326** (20087) and **TVBE-51320** (20144) [author photo — plates C-36, C-39].

The labels are transcribed here and nothing is inferred from them. No claim is made about the organization, its facilities, its broadcast activities, when it held the units, or how they left its custody; the project has sought no documentation on any of these points and the research programme is closed (Appendix F). What the labels establish, and all they establish, is that **both export units passed through one institution’s asset-tracking system**, and that their tag numbers are six apart — the same evidentiary shape as the consecutive UAE seal serials of §4.4, and the second instance in this project of two units of one purchase carrying near-adjacent third-party identifiers. The tags are the only provenance either export unit carries.

E.3 Physical description

Format. 2RU rack unit — twice the AMS-100's height — with two internal speakers behind metal grilles flanking a central meter block, and a measured mass of **35 lb (≈ 15.9 kg)** on the export units [author, measured]. This is the project's only weight figure for any product of the family: the corresponding AMS-100 measurement is declined by standing ruling (A.1), and the two figures should not be read against one another as a like-for-like comparison. The ソニー株式会社 plate text and 100 V-only rating mark the contributor units as Japan-domestic; the author's units are marked for four supply voltages (E.2). On both variants the 150 W rating is matched by a prominent finned heatsink running the chassis side [contributor photo, EXT-01]: a substantially larger amplifier stage than the successor's (the AMS-100 draws 25 W [IM]), and the readiest explanation of the mass.

Front panel. POWER rocker with separate indicator window; two illuminated **analog VU meters** (-20 to $+3$ scale with a percentage-modulation scale beneath, red segment above 0) labeled INPUT LEVEL L/R; two rotary **SP LEVEL** pots (0-10 scale), one per channel; the legend AUDIO MONITOR SPEAKER AMS-3 across the meter block. No input selection of any kind — what the rear receives is what the speakers monitor. Panel finish on the export units is light grey [author photo, plate C-37].

Rear panel. Fixed, analog-only I/O, identical in complement across both variants:

Facility	Detail
Inputs, per channel	XLR-32F (male) + XLR-31F (female) pair, ITT Cannon — true loop-through
Input impedance, per channel	slide switch: 600 Ω / 10 kΩ / HIGH
Input level, per channel	slide switch: 8 dB / 4 dB / 0 dB
Speaker routing	SPEAKER SELECT slide: INT. / EXT.
External speakers	EXT SPEAKER R/L XLR pair, 8 Ω
Other	ground post; IEC AC inlet; one covered rotary component (see below)

Rear-panel readings for the domestic pair [contributor photo, EXT-01]; connector types and switch legends corroborated at author tier on the export pair [author photo, plate C-38]. Switch functions are read from panel legends, not verified electrically.

One unresolved fitting. A large rotary component under a clear protective cover sits at the right of the rear panel on all four units. It was recorded as a fuse holder when only the domestic pair was documented. On a chassis marked for four supply voltages, a covered **voltage selector** is at least as likely, and the protective cover is more characteristic of a selector than of a fuse; the legend beneath the cover is not readable in any photograph in hand. **The identification is left open** [candidate reading, flagged not ruled — no examination was made, and none is scheduled (Appendix F)]. If it is a selector, its presence on the domestic 100 V-only units would itself be worth knowing, and nothing in the record settles that either.

E.4 The comparative reading — design vocabulary across a generation

Laid against the AMS-100 (Sections 2-3), the AMS-3 reads as the same *monitoring concept* executed a hardware generation earlier. The continuities and the breaks are both instructive; all of the following is comparative observation from documented hardware on each side, not lineage evidence:

What carried forward. The per-channel input-conditioning grammar is recognizably continuous: switched impedance including a 600 Ω termination position, switched input level, XLR loop-through per channel. The AMS-100's BKAM-101 analog board speaks a terser dialect of the same language — 600 Ω termination on/off, +4/0 dB level [IM][MM] — where the AMS-3 offers three positions on each axis (600 Ω / 10 k Ω / HIGH; 8/4/0 dB). The loop-through convention survives the generation intact, reappearing on the 101's connector pairs [MM] and even on the 103's SDI BNC pair.

What the redesign changed. Three moves, each in the same direction — from a self-contained analog monitor toward a compact, system-integrated one:

1. **Fixed I/O → modular slots.** The AMS-3's hardwired analog inputs became the AMS-100's two-slot option architecture (Section 3) — the defining change, and the one that makes the successor's configuration a documentable variable at all.
2. **Analog VU → LED bar metering,** with the AMS-100 adding front-panel input selection and calibration modes the AMS-3 has no equivalent for.
3. **On-board external-speaker facility dropped.** The AMS-3 drives external 8 Ω speakers directly, with front-panel INT/EXT routing; the AMS-100 has no frame-level equivalent — amplified line-out moved to the REMOTE connector's monitor-output plane [IM], trading the facility for the 1RU height and the smaller amplifier.

A fourth move, withdrawn. Earlier revisions of this appendix listed a fourth: *Japan-domestic 100 V → universal 100-240 V supply*, read as evidence that the successor was built for worldwide deployment and the predecessor candidate was not. **That reading is withdrawn.** It rested on the two domestic units then in the record, and the appendix flagged its own exposure at the time — whether export-market AMS-3 variants existed was stated as unknown. The author's units answer it: they exist, and the AMS-3 was sold into multiple mains markets.

What survives is narrower, and is a statement about supply *architecture* rather than market ambition: **tap-selected multi-voltage (100/120/220/240 V) → continuous auto-ranging universal supply (100-240 V)** [E.2 vs. IM]. The predecessor candidate reached export markets by offering four discrete voltage settings, presumably selected at installation; the successor reached them with one supply that requires no setting at all. That is a real generational change in how a product is made deployable, and it is a smaller claim than the one it replaces. The withdrawal is recorded rather than quietly corrected because the original reading was published, and because it is a clean instance of what this monograph's evidence tiers exist to catch: a confident generalization drawn from $n = 2$ of one variant, falsified by the next two units to enter the record.

Weight of the comparison. The continuity of input-conditioning vocabulary is real and specific — 600 Ω switched termination on a rack confidence monitor is not so

generic a feature that its recurrence means nothing. But convergent design within one manufacturer's pro-audio group explains it just as well as direct succession. The comparison makes the lineage *plausible*; it cannot make it *sourced*. [author reading of the comparative evidence; moderate for the continuity observation itself, no upgrade to the lineage claim]

E.5 The lineage question — state and paths

State: unverified, low confidence, unchanged (§1.4). Two documentary paths have now returned nothing: period paperwork with the contributor units, and period paperwork with the author's own two — neither group arrived with a manual, a carton, or a document of any kind. Four units, no paper. The paths that could in principle move the question are recorded for completeness and are not pursued by this project (Appendix F): a listing carrying AMS-3 manuals, brochures, or boxes in any market [no claim is made about how often such listings appear]; the AMS-100 Operation Manual, not located (§2.8); period Sony broadcast catalogs of the mid-1990s.

That four physically documented units move the lineage claim not at all is the point worth stating plainly. Each new unit sharpens the description of the product and can falsify a comparative reading — as the export pair has just done to E.4's fourth move — while leaving the succession question exactly where marketing copy left it. The formulation of record stands: **the AMS-3 is a documented product and a credible predecessor candidate; the succession itself remains marketing copy.**

Sources: EXT-01 photo packs, July 2026 (22 images; intake record in the project file) [contributor photo, EXT-01] — plates C-19-C-21; the author's photographs and handling of AMS-3 units 20087 and 20144, August 2026 [author; author photo] — plates C-35-C-39, weight author-measured; Sony Installation Manual [IM] and Maintenance Manual [MM] for all AMS-100-side comparanda. Contributor credit: Appendix D. Bounds for this appendix are stated in Appendix F.

Appendix F — Bounds of the Record

This appendix states where the monograph's record stops. The research programme behind it closed in August 2026 by author ruling: no further acquisition, inquiry, recovery, measurement, or observation is scheduled, and nothing below is an open task. Each line names a question the record leaves unanswered, the publication home of the finding it bounds, and the ruling that closes it, so that no reader, contributor, or later editor mistakes a gap for an oversight or reopens a question that was closed on purpose. Frames are identified by serial throughout; the drafting record numbers (R-1 ... R-7) map to frame serials at Appendix D, D.3, and nowhere else; the author's private record identifiers are not published. This appendix records; it does not decide, rank, or resolve.

Documentation

- **Operation Manual — not located; search closed.** The third Sony document of the set (§2.8) was confirmed absent from the online manual aggregators at the July 2026 literature sweep, and the physical search is closed. Every figure and statement that only it could have verified stands permanently at its current tier: the dealer-only frame specifications (speakers, headphone output, monitor-output maximum and impedance — §2.4, A.1–A.2); the dealer-only board figures (BKAM-101 maximum level and bridging impedance, BKAM-102 BNC impedance — §3.2, §3.3, A.3–A.4); frame weight and depth (A.1 — measurement declined by ruling); the operator-facing wording of the BKAM-103 channel-select procedure (§3.4) and the behaviour of the GP/CH buttons with a BKAM-102 selected (§5.6 — not investigated by experiment); the per-line REM/STS-to-slot mapping and the extent of the level/pulse type beyond pins 14/15 (Appendix B); the AMS-3 lineage (E.5). The D.1 entry is closed as not located.
- **Literature claim is dated.** The statement that no editorial or community literature on the AMS-100 exists (§1.1, §7.1) is a finding of the July 2026 sweep; no periodic re-sweep is scheduled, and the claim publishes as dated, not as standing.
- **Period Sony literature.** The BVM/PVM pairing claim (§1.3) rests on marketing copy, and the AMS-3 lineage (§1.4, Appendix E) stays at low confidence by design; neither a period catalog nor a price list was sought beyond the July 2026 sweep. No dealer inquiry is made for the Operation Manual, loose-board restock, or AMS-3 paperwork.

Hardware

- **The AMS-3 export pair — documented, not investigated.** Units 20087 and 20144 are described in Appendix E from the author’s photographs and handling. Three things about them are recorded and not pursued: the identification of the covered rotary component on the rear panel, left open between a fuse holder and a voltage selector (§E.3) — no examination is scheduled; the institutional asset tags on both units, transcribed and nothing more (§E.2.1) — no enquiry is made of the named organization, and no claim about it appears anywhere in this monograph; and the significance of the 2xxxx serial band, on which four units support no inference (§E.2). AMS-3 20087 has been sold. Neither unit is a registry member and neither generates a Section 7 ledger row.
- **A published reading withdrawn.** Appendix E’s former fourth redesign move — Japan-domestic supply giving way to a universal one across the generation — was falsified by the export pair and is withdrawn in place rather than deleted (§E.4). It is recorded here as the record’s clearest instance of a generalization drawn from one variant at $n = 2$ failing against new hardware; the surviving claim is narrower and is about supply architecture only.
- **Registry board-side recheck: declined.** The five-digit serial status of the BKAM-103 labels 0105241 / 0128521 and of 12793’s BKAM-102 is permanently unknown. No board is pulled from a working installation for identification, dating, QFP confirmation, or serial reading alone; the BKAM-103 large QFPs

(CXB1342R / CXD8280AQ) stay at Maintenance-Manual tier, and the serials of the four Japan-recycler BKAM-101s arriving later are unread (§4.2, §4.4, §6.4).

- **BKAM-102 s/n 10083 — never read, now installed.** Its serial is assigned by elimination against carton S01-0010083 and was never read from the PCB. The board was installed into either frame 12886 or frame 13048 without being read, and 13048 has since been sold, so the carton-to-board mapping **closes permanently at two of three** (§4.2). Which of the two frames received it, and which received 12793's as-surfaced BKAM-102, is unstated and stays so.
- **Placements and states not recorded.** The individual routing of the four unread BKAM-101s among 10010, 11341, 12793, and 12886, and 11341's slot letters — the census fixes the totals and the origins, not which board sits where (§4.2); the serials of the boards in 12886 and 13048, never read at any point; the cabling of every BKAM-101-fed monitor point, the disposition of the converters, and the splitter state at 12793 at its conversion to analog (§5.2, §5.3); the S101 position of the one installed BKAM-102 (§5.6); the installation role of frame 12886 (§5.3); the 600 Ω as-found switch positions of every BKAM-101 handled (§3.2); the STS level/pulse configuration of every frame (§6.2); the CRT-top placement specifics (§5.5). All are undocumented and stay so.
- **Seal C094182 on frame 10010.** In place as last checked at the author's rebuild; not re-verified since (§4.3, §4.4). Chassis sweeps were not run on 11878, 12886, or 13048 (§4.4).
- **Date-code candidates: not adopted.** Two 2000-family board marks (a five-character mark on a BKAM-101, a six-digit mark on a BKAM-103) stay in the project record; no code-format ruling or board sweep is made. The production window publishes as "at least into 1999" and the end of production as unknown (§1.5, §6.4). Sony external marking for this family is confirmed date-free (seven registry plates, three cartons, the contributor's plate, and all four AMS-3 plates across both market variants).
- **§6.4 discrepancies stand as recorded.** The BKAM-102 CPLD/firmware readings (EPM7064S V1.10 on the contributor's s/n 10016; EPM7032S V1.20 on the author's s/n 10099) and the BKAM-103 per-CPLD firmware readings (V1.00/V1.00 on s/n 10130; V1.10/V1.00 on s/n 10631) are flagged, not reconciled, and no re-verification is made; neither value is tabulated as a family specification (A.4, A.5).
- **HD-SDI option board BKAM-101 (§3.1).** Attested by one social-media photograph; part number unread; untabulated in Appendix A and outside the Maintenance Manual's coverage statements. EXT-03 is anonymous, uncontacted by ruling, photo-tier only, and closed at opening; its screenshots are not plates.

Images

- **No image is reproduced.** Every plate in Appendix C publishes as a caption. The twenty author photographs (C-1–C-9, C-24–C-34) are withdrawn from publication and held in the project record; every reading that rested on them stands

at [author] tier on that record. The nine EXT-01 images in the project file and the two archived seller-listing files are held, not embedded.

- **C-11 / C-13 / C-18.** Never supplied by the contributor; recovery closed. The BKAM-103 firmware divergence that C-13 would have re-checked stands as a flagged discrepancy (above).
- **C-23 custody.** The second seller-listing archive filename’s hash differs from the project’s capture record, which lists a “front” as the second file; five distinct files are held against six listing images. Both discrepancies are recorded and stand unreconciled (Appendix C, D.3).

Market and transactions

- **Author transactions: withheld, not unknown.** No author figure, band, approximation, direction, or ratio publishes; no author transaction date publishes beyond the year; sale particulars beyond event and configuration do not publish; the author’s sources publish as exactly three channel classes — a UAE electronics recycler, a Japan electronics recycler, and a US electronics recycler (the sealed BKAM-102 lot) — with no finer characterization; counterparties, dealers, and venues of the author’s own transactions are unnamed (§1.1, sole statement). Whether any registry frame is offered for sale at any time is not published; registry status is held or sold only (§4.1). The author’s private record identifiers are not published for any frame (D.3).
- **Third-party prices as observed.** The \$999.99 US-visible frame ask (§7.2, §7.3) and a Tokyo dealer’s ¥13,200 sold-out loose BKAM-101 listing are July 2026 observations; neither is re-observed, no currency conversion or rounding of either is published, and the ≈2017 US hi-fi reseller sale stays an existence datum with no price (no seller inquiry is made).
- **No rate claims, ever.** Nothing in the record supports a statement of how often AMS-100s or BKAM boards surface in any channel; every population statement is a dated, point-in-time observation (§1.7, §7.1, §7.8). The daily-search observation log is retired and is never cited. No seal-recurrence, loose-board-supply, or per-board BKAM-101 market watch is maintained.
- **EXT-01 and EXT-02: no follow-up, no ledger rows.** EXT-01’s contribution is closed at the credit line of record (Appendix D); provenance beyond custody geography is skipped permanently, and no ledger row is derived from it. EXT-02 stays at report tier — one Los Angeles unit with a reported BKAM-101, non-working, retained by its owner — with its withdrawn ask and failure symptoms unrecorded; no photo pack is sought (§4.5.2).

Labels and record artifacts

- **Compliance and aftermarket labels: out of publication.** The top-cover compliance label present on all seven frames was removed by the author; its partial reading from a listing photograph is the permanent basis and is not published. The small aftermarket label on 11341 is not for discussion.
- **Name-implied is not data.** REM-A / STS-A → SLOT A and REM-B / STS-B → SLOT B is a naming inference (Appendix B); the select-code and pin tables are verbatim transcriptions and carry no interpretation.

- **Records are outside this monograph.** Amendment of the author's private records for the first three frames, identifiers for the four later frames, the sale record, and the region-field reconciliation are matters of the author's private record and have no publication consequence.

Sources: the author's rulings of July-August 2026 closing the research programme; the open-item lines formerly carried by Sections 1-7 and Appendices A-E, as consolidated and closed; Sony Installation Manual [IM] and Maintenance Manual [MM] for the manual-coverage statements cited; Appendix D for every record referenced.